

Screening Two Types*

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Abstract

We characterize profit-maximizing menus in screening settings with an arbitrary set of alternatives in which the agent has one of two privately-known types. We assume that utilities are quasi-linear but impose no other restrictions (such as increasing differences) on the agent's utility or the set of alternatives. Our characterization relies on a novel geometric approach, clarifies the role of increasing differences in the standard setting, and shows when random menus are beneficial. We describe novel applications to vertical and horizontal differentiation and multi-product bundling.

1 Introduction

Screening settings, in which a principal faces a privately-informed agent, have been studied extensively in the economics literature, beginning with the seminal contributions of Mussa and Rosen (1978), Riley and Zeckhauser (1983), and Maskin and Riley (1984). The principal designs a menu of (alternative, price) pairs to maximize his profit, and the agent maximizes his quasi-linear utility by choosing either an optimal

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pair from the menu or his outside option. The agent’s private information (his type) determines his utility from the outside option and the various alternatives.

A central challenge in analyzing such settings is that the alternatives are often “multi-dimensional.” In a price discrimination setting, products may have multiple attributes, which different buyer types may rank differently; in a bundling setting, different buyer types may rank products differently, leading to different rankings over bundles. This makes the analysis difficult because it is not a priori clear which incentive constraints optimally bind. Much of the literature has instead focused on single-dimensional models in which all types have a common ranking of the alternatives and a higher type is willing to pay more than a lower type to obtain a better alternative. A typical price discrimination model, for example, assumes that products are ranked in terms of quality and higher types are willing to pay more for a quality increase. Such “standard” settings lead to beautiful insights (such as “efficiency at the top”) and useful results, but they limit the applications that can be studied, and it is not clear to what degree the resulting important insights on the economics of screening apply without the additional structure.¹

This paper analyzes screening settings in which the set of alternatives is unrestricted and the agent has one of two types. In particular, the alternatives may be complex and need not be ranked, and neither type is necessarily “higher” than the other. We obtain a complete characterization of the optimal (profit-maximizing) menus by employing a novel yet simple approach, which avoids the need for a complicated “case-by-case” analysis.² This provides a unified framework for solving two-type screening models and, as we illustrate below, expands the set of applications that can be investigated. The characterization also clarifies which features of the optimal menus in two-type environments arise from the economics of screening and which features in the standard setting rely on its additional structure. Additionally, the characterization shows when random menus increase the principal’s profit (they

¹Solutions to specific questions in certain non-standard environments have been obtained (for example, Haghpahan and Hartline, 2021; Ghili, 2023; Yang, 2025); their specificity testifies to the difficulty of studying screening in general.

²Many screening settings, including ours, can be written as linear programs. One can then consider every possible combination of binding incentive constraints, guess Lagrangian weights for those constraints, and verify optimality using those guesses. Even with two types, there are many cases to consider (in principle any subset of the four constraints might bind). Moreover, such an approach would not explain why certain constraints bind or are slack, so it provides little economic insight. Our technique avoids this guess-and-verify approach and clarifies the underlying economic forces.

do not in the standard setting).

Before describing our novel approach, we comment on why the two-type model merits an in-depth analysis. First, the two-type model is frequently the first model studied when novel applications or settings are explored. And many papers in both mechanism design and game theory focus entirely on two types.³ Possibly for this reason the two-type model is often the first screening model taught, which may be another reason to try to understand it completely; it is perhaps surprising that such a treatment has not yet been conducted. Moreover, the unrestricted set of alternatives, which can be infinitely dimensional (representing, for example, experiments, segmentations, or databases, as in Bergemann et al., 2018, Yang, 2022, and Eliaz and Spiegler, 2025) makes the model maximally flexible in terms of the alternatives it can represent. Second, many real-world settings (with possibly complex alternatives) are inherently two-type: the agent may be guilty or not guilty of a crime, able or not able to complete a task, informed or uninformed, rational or behavioral, good or bad, willing or not willing, etc. Third, as mentioned above, the solutions to screening models with additional restrictions are a joint implication of the economics of screening and the additional restrictions. By contrast, our treatment uncovers fundamental economic properties in two-type environments that emerge without additional restrictions. Finally, as the price discrimination and bundling applications with unordered alternatives below demonstrate, our framework makes new settings amenable to analysis, including settings that have so far been difficult to analyze.⁴

Our approach to characterizing the optimal menus includes several key ideas. The first innovation is to work in the space of valuation pairs, in which each alternative a is represented by the pair $(v_1(a), v_2(a))$, where $v_i(a)$ is type i 's valuation for alternative a net of production costs. The space of valuation pairs lies in $\mathbb{R}^2$ regardless of the dimensionality or complexity of the set of alternatives, and can be represented geometrically as in Figure 1. This facilitates a geometric approach to the characterization.

The second innovation is the notion of an *uncontested* type. We say that a type

³Even papers that focus on a continuum of types often include a treatment of two types (Frick et al., 2025 is one recent example).

⁴Evan in settings that naturally involves more than two types, a two-type model may be a valuable first step that provides initial economic insights and indicates whether a more elaborate investigation is warranted. And when such an investigation proves difficult, solving a two-type model may be preferable to abandoning the exercise entirely.

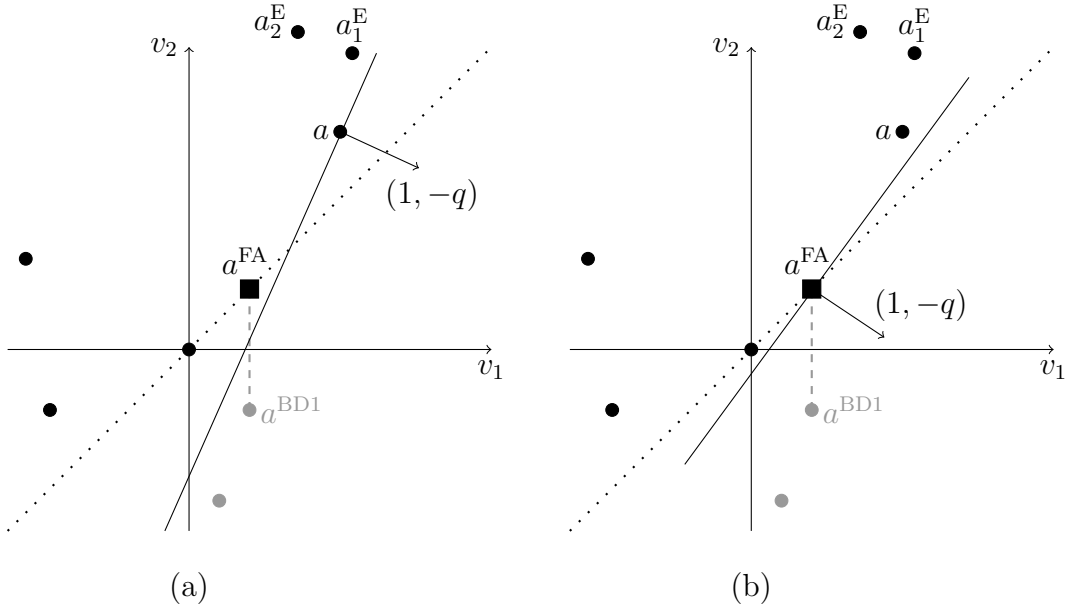


Figure 1: Each alternative is represented by a circle. The efficient alternatives are a_1^E and a_2^E , type 1 is contested, and type 2 is uncontested. The fictitious alternative, represented by a square, is the projection of a^{BD1} onto the diagonal. (a) The maximal alternative a is not the fictitious alternative, type 1's allocation is alternative a , and type 2's IC constraint binds. (b) The fictitious alternative is maximal, type 1's allocation is alternative a^{BD1} , and type 2's IR constraint binds.

is uncontested if his valuation for his efficient alternative weakly exceeds that of the other type for that alternative; otherwise the type is *contested*. In particular, whether a type is uncontested is immediate from the types' valuations for the efficient alternatives and does not depend on the proportions of the types or their valuations for any of the other alternatives. In Figure 1, type 1 is contested and type 2 is uncontested.

We show that at least one type is uncontested, and our key “No-Tradeoff Lemma” shows that in any optimal menu an uncontested type is served efficiently and the other type obtains utility zero. In particular, the “first-best” is implementable if and only if both types are uncontested.⁵ An uncontested type generalizes the notion of a “high type” from the standard setting;⁶ but unlike in the standard setting, in an optimal menu an uncontested type's incentive constraint may be slack, and his individual

⁵The “first-best” refers to selling each type his efficient alternative for a price equal to his valuation for that alternative.

⁶If both types are uncontested, then they are both “high.”

rationality constraint may bind (so he obtains utility zero) even if the other type is not excluded. If only one type is uncontested, then which of his two constraints bind, together with the allocation of the other (contested) type, pins down the optimal menu.

Our third innovation is to geometrically identify, in the space of valuation pairs, which constraint(s) of the uncontested type binds and the allocation of the contested type. We first consider the 45 degree line through the origin (“the diagonal”) and identify the alternative a^{BD1} that the contested type, without loss of generality type 1, values the most among the ones weakly below the diagonal.⁷ We then project a^{BD1} on the diagonal to obtain a “fictitious alternative” a^{FA} that both types value equally (see Figure 1). We then find an alternative among the ones weakly above the diagonal that is maximal in the direction of the vector $(1, -q)$, where q is the proportion of the uncontested type (type 2). If this maximal alternative is not the fictitious alternative, then this maximal alternative is allocated to the contested type and the uncontested type’s IC constraint binds. If it is the fictitious alternative, then the alternative a^{BD1} used for the projection is allocated to the contested type and the uncontested type’s IR constraint binds. Figure 1 (a) illustrates the former case and Figure 1 (b) illustrates the latter case. The geometric approach also allows us to characterize when offering a random alternative to the contested type increases the principal’s profit. A sufficient condition (for high enough q) is that the types’ valuations for alternative a^{BD1} differ, as is the case in Figure 1.⁸ An example in Appendix B illustrates our characterization of the optimal menus, how they change with the proportion of the types, and the role of random alternatives.

Beyond providing a unified framework for solving two-type screening models, our characterization makes new settings amenable to analysis. We demonstrate this with two applications. The first combines vertical and horizontal product differentiation and assumes that the cost of production is linear in a product’s vertical value (and independent of its horizontal value). Both types are uncontested, so first-best is implementable, if the types’ ideal horizontal values are sufficiently different given the types’ preference intensities. When this is not the case, we show that the horizontal value of the contested type’s allocation is optimally distorted away from both types’

⁷BD1 stands for “best alternative below the diagonal for type 1.”

⁸The necessary and sufficient condition is that convexifying the set of valuation pairs leads to a fictitious alternative for this expanded set that lies strictly higher on the 45 degree line through the origin than the fictitious alternative for the original set of valuation pairs.

ideal horizontal values, and the distortion increases, up to a point, in the probability of the uncontested type. The vertical value of the contested type’s allocation is not distorted unless the contested type’s preference intensity and probability are sufficiently low. In that case, the contested type is optimally excluded and only the uncontested type is served.⁹

Our second application is to bundling. We show that when one of the types is contested, his allocation can optimally be distorted in two ways. First, some products may be removed from the contested type’s efficient bundle if they are valued sufficiently highly by the uncontested type. Second, and perhaps unexpectedly, some products that *both* types dislike may be added to the contested type’s bundle if the uncontested type dislikes these products sufficiently more than the contested type. These distortions increase, up to a point, in the probability of the uncontested type.

The rest of the paper is organized as follows. Section 1.1 surveys the related literature. Section 2 describes the model and preliminaries. Section 2.1 reviews the standard setting with increasing differences. Section 3 analyzes the model, provides the main result, and describes the main parts of the proof. Section 4 considers the case in which the set of valuation pairs is convex, and characterizes when offering random alternatives increases the principal’s profit. Section 5 describes the product differentiation and bundling applications. Section 6 revisits the standard setting with increasing differences and concludes. The appendix completes the proof of our main result. The supplemental appendix contains an illustrative example, details of the cost and valuations normalization, an expanded discussion of increasing differences, the case of a convex set of valuation pairs, the use of random alternatives, and proofs of auxiliary results.

1.1 Related literature

Papers that study screening often focus (entirely or partially) on two-type versions of the problems they study. Bergemann et al. (2018) and Boleslavsky and Kolb (2025) study selling information structures, Rochet and Stole (2002) study random participation constraints, Pycia (2006) studies bundling products with additive values, Galperti

⁹When production is costless, the contested type is only excluded if his probability is sufficiently low and the two types’ ideal horizontal values are identical. In this sense, the result in the standard price discrimination setting with costless production that the low type is excluded when his probability is sufficiently low is non-generic.

(2015) studies screening an agent with self-control problems, and Eliaz and Spiegler (2025) study designing databases. Using our result, Bergemann et al. (2025) study a two-type version of the problem of designing large language models. These applications impose additional structure on types' utilities beyond quasi-linearity, and our paper clarifies which properties of optimal menus hold generally and (consequently) which ones rely on the additional structure.

Turning to our applications, the classical models of monopolistic price discrimination (Mussa and Rosen, 1978 and Maskin and Riley, 1984) allow only for vertically differentiated products and assume increasing differences. In these models, the only possible distortion in a type's allocation is a downward distortion. Our model makes it possible to study the *direction* of the distortion. Our application to vertical and horizontal differentiation shows that the horizontal value of the contested type's allocation may be distorted away from the ideal horizontal value of the uncontested type, and our application to bundling shows that products with a negative valuation might be *added* to the contested type's bundle.

The literature that studies horizontal differentiation of products mostly focuses on imperfect competition, where each firm sells one of the products. The classical models (Hotelling, 1929 and Salop, 1979) allow only for horizontal differentiation (and not vertical differentiation). Villas-Boas (1999) and Armstrong and Vickers (2001) add horizontal differentiation to the vertical differentiation model of Mussa and Rosen (1978) in discrete-choice models in which a random additive shock is added to the value of each product for each consumer. We are not aware of papers that combine horizontal and vertical differentiation in a monopoly setting, as in our application.

Our bundling application is related to the literature on multi-product bundling that started with Stigler (1963) and Adams and Yellen (1976). The theoretical literature highlights that optimal menus might be complex. Optimal menus may be random (Thanassoulis, 2004), include an infinite number of alternatives, (Vincent and Manelli, 2007), and be hard to compute (Daskalakis et al., 2014). As a result, large parts of the literature impose additional structure on the problem, and solutions are known only in very special cases. Even in what is considered the simplest case, two products with additive and independently drawn values, optimal menus can still be complex and are identified only for specific examples (Daskalakis et al., 2014, Thirumulanathan et al., 2019). We simplify the problem in another direction: instead of assuming that there are two products (with independent values), we assume that

there are two types, allow for any number of products and any quasi-linear valuations, and fully solve the problem.¹⁰

2 Model and preliminaries

A profit-maximizing principal faces an agent whose privately-known type $t \in T = \{1, 2\}$ is 1 with probability $1 - q$ and 2 with probability $q \in (0, 1)$. There is a set A of alternatives, where alternative $0 \in A$ represents the agent's outside option. An alternative can correspond to a quality/quantity of a product, to a bundle of products, or to a list of characteristics of a product. The agent's type t specifies his valuation $v_t(a) \in \mathbb{R}$ for every alternative a . Type t 's utility from alternative a and payment p to the principal is $v_t(a) - p$. The principal's cost of producing alternative a is $c(a) \in \mathbb{R}$.

Denote by $V = \{(v_1(a), v_2(a)) : a \in A\}$ the set of valuation pairs across the alternatives and assume that V is closed. Let

$$a_t^E = \arg \max_{a \in A} (v_t(a) - c(a)) \quad (\text{E1-E2})$$

be the efficient alternative of type t , and assume that it exists and is unique.¹¹ Closedness of V and existence of a_t^E are satisfied for example when A is finite or, more generally, when A is compact and v_t and c are continuous.

The principal designs a menu of (a, p) pairs, where $a \in A$ is an alternative and p is a price. The agent chooses either a pair from the menu or his outside option to maximize his utility. Notice that we require the principal to offer only deterministic alternatives. This is more general than allowing the principal to offer random alternatives, because we can consider random alternatives by convexifying V . We discuss random alternatives in Section 4.

By the revelation principle (Myerson, 1979), and because the principal can include the outside option with a price of zero in the menu, it suffices to consider

¹⁰Our main result imposes no structure on types' valuations and can be used to identify optimal menus without assuming additivity across products. We impose additivity in the bundling application for two reasons. First, additivity is a canonical assumption made by most of the literature (with exceptions that include (Haghpanah and Hartline, 2021, Ghili, 2023, and Yang, 2025)). Second, with additivity we can describe the solution geometrically.

¹¹Existence is important for the existence of optimal menus. Uniqueness is for expositional simplicity only.

menus $\{(a_1, p_1), (a_2, p_2)\}$ from which each type t chooses the pair (a_t, p_t) .¹² Thus, the principal solves the following problem:

$$\begin{aligned}
& \max_{a_1, a_2 \in A \quad p_1, p_2 \in \mathbb{R}} (1 - q)(p_1 - c(a_1)) + q(p_2 - c(a_2)) \\
& \text{subject to} \quad v_1(a_1) - p_1 \geq v_1(0), \\
& \quad \quad \quad v_2(a_2) - p_2 \geq v_2(0), \\
& \quad \quad \quad v_1(a_1) - p_1 \geq v_1(a_2) - p_2, \\
& \quad \quad \quad v_2(a_2) - p_2 \geq v_2(a_1) - p_1.
\end{aligned}$$

Appendix C shows that we can normalize the production costs of all the alternatives and the valuations for the outside option to zero, which may result in negative valuations for some alternatives, in a way that maintains the same solutions to the principal's problem.¹³ Consequently, we will solve the following problem:

$$\begin{aligned}
& \max_{a_1, a_2 \in A \quad p_1, p_2 \in \mathbb{R}} (1 - q)p_1 + qp_2 \\
& \text{subject to} \quad v_1(a_1) - p_1 \geq 0, & \text{(IR1)} \\
& \quad \quad \quad v_2(a_2) - p_2 \geq 0, & \text{(IR2)} \\
& \quad \quad \quad v_1(a_1) - p_1 \geq v_1(a_2) - p_2, & \text{(IC1)} \\
& \quad \quad \quad v_2(a_2) - p_2 \geq v_2(a_1) - p_1. & \text{(IC2)}
\end{aligned}$$

Figure 2 depicts an example of the normalized set V , which may contain valuation pairs with positive or negative values, and highlights the valuation pairs of the efficient alternatives a_1^E and a_2^E , which are the rightmost alternative and the topmost alternative, respectively. Throughout the paper, we sometimes identify an alternative a with its associated valuation pairs $(v_1(a), v_2(a))$ when this should not cause confusion. For example, we can say that in Figure 2 alternatives a_1^E and a_2^E lie in the positive orthant.

If the principal could observe the agent's type, that is, if we ignored the constraints IC1 and IC2, then the solution would be the "first-best," which assigns to each type his efficient alternative at a price equal to his valuation for that alternative. If the

¹²It may be that $a_1 = a_2$.

¹³We change type t 's valuation for every alternative a to $v_t(a) - c(a) - (v_t(0) - c(0))$. This does not change the efficient alternatives a_1^E and a_2^E .

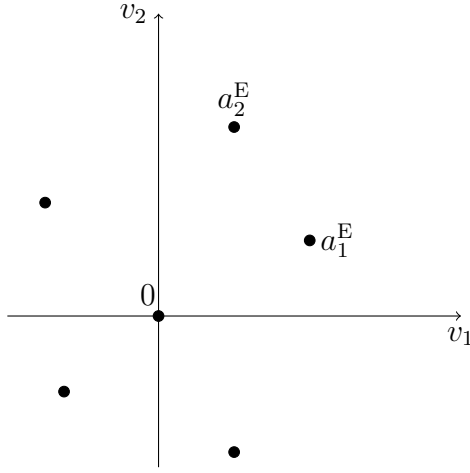


Figure 2: The normalized set V may contain valuation pairs with positive or negative values.

first-best also satisfies IC1 and IC2, then this is the solution to the principal's problem with all the constraints. We then say that the optimal menu implements the first-best, or that the first-best is implementable.

Definition 1 *A menu $\{(a_1, p_1), (a_2, p_2)\}$ implements the first-best if $a_t = a_t^E$ and $p_t = v_t(a_t^E)$ for $t = 1, 2$.*

Our main result solves the principal's problem with all the constraints and also shows when the solution implements the first-best.

2.1 Increasing Differences

The screening literature usually assumes that the alternatives are ordered, with the outside option being the lowest alternative, and that one of the types, say type 2, values any increase in his allocated alternative strictly more than the other type, type 1. This “increasing differences” condition facilitates the characterization of the optimal menus because it allows us to ignore IR2, which then makes it possible to repeatedly simplify the problem and obtain a formulation that is separable across a_1 and a_2 . This separable formulation implies that in any optimal menu type 2 (the “high type”) is served efficiently (“efficiency at the top”), that IR1 and IC2 hold as equalities, and that the allocation of type 1 (the “low type”) may be distorted downward to reduce type 2's information rent. As the proportion q of type 2 approaches 1, type 1 is optimally excluded or “almost excluded” in that in the space V of valuation pairs type 1's

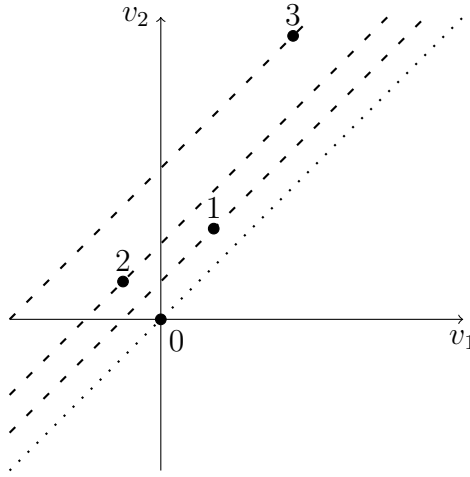


Figure 3: An example with alternatives $\{0, 1, 2, 3\}$ that satisfies increasing differences.

allocation approaches the outside option. In addition, the first-best is implementable if and only if it is efficient to exclude type 1. We formally review these results, as well as a weaker notion of increasing differences, in Appendix D, and highlight that these results crucially rely on the outside option being the lowest alternative. In the space of valuation pairs, with increasing differences all the alternatives lie above the 45 degree line through the origin, and higher alternatives lie on higher 45 degree lines, as illustrated in Figure 3.

Without increasing differences, alternatives can lie anywhere in the space of valuation pairs. There are no natural “high” and “low” types, distortions can take various forms, and no constraint can be ignored in order to simplify the problem and ultimately obtain a separable formulation. In particular, it is not clear which constraints hold as equalities at the optimum. Characterizing the optimal menus requires a different approach, which we now present.

3 Characterizing Optimal Menus

Theorem 1 below characterizes the optimal menu. The characterization hinges on the location of the efficient alternatives a_1^E and a_2^E relative to the 45 degree line through the outside option $(0, 0)$. We will refer to this line as “the diagonal.” Alternatives that lie above the diagonal are those for which type 2’s valuation exceeds that of type 1, and the reverse holds for alternatives that lie below the diagonal.

Whether a type's efficient alternative lies above or below the diagonal corresponds to whether the type is *uncontested*, as given by the following definition.

Definition 2 *Type t is uncontested if $v_t(a_t^E) \geq v_{t'}(a_t^E)$, where $t' \neq t$. Otherwise, type t is contested.*

An uncontested type is willing to pay weakly more for his efficient alternative than the other type does.¹⁴ Type 1 is uncontested if and only if a_1^E lies weakly below the diagonal; type 2 is uncontested if and only if a_2^E lies weakly above the diagonal. In particular, whether a type is uncontested depends only on the types' valuations for the efficient alternatives and is independent of the types' probabilities or their valuations for any of the other alternatives.

We observe that at least one type is uncontested. This is because alternative a_2^E lies on a weakly higher 45 degree line than alternative a_1^E , that is, $v_2(a_2^E) - v_1(a_2^E) \geq v_2(a_1^E) - v_1(a_1^E)$.¹⁵ Thus, there are three possibilities, depicted in Figure 4: (a) Both types are uncontested (a_2^E lies weakly above the diagonal and a_1^E lies weakly below the diagonal), (b) type 1 is contested and type 2 is uncontested (both alternatives lie strictly above the diagonal), or (c) type 1 is uncontested and type 2 is contested (both alternatives lie strictly below the diagonal). In cases (a) and (c), type 1 is uncontested; in cases (a) and (b), type 2 is uncontested. Without loss of generality, we assume throughout our analysis that type 2 is uncontested. That is, if we are in case (c), switch the labels of the types to obtain case (b).

Assumption 1 *Without loss of generality, type 2 is uncontested ($v_2(a_2^E) \geq v_1(a_2^E)$). That is, either Figure 4, (a) or Figure 4, (b) applies.*

3.1 No Tradeoff Lemma

Uncontested types play a crucial role in our characterization of the optimal menus. This is because, as the following key lemma shows, the principal does not face a trade-off between maximizing the surplus that can be extracted from an uncontested type and minimizing the information rent of the other type, regardless of the probabilities of the two types.

¹⁴Thus, An uncontested type would outbid the other type in a standard auction for the uncontested type's efficient alternative, which motivates the term "uncontested."

¹⁵This follows because, by definition of a_1^E and a_2^E , $v_2(a_2^E) \geq v_2(a_1^E)$ and $v_1(a_1^E) \geq v_1(a_2^E)$.

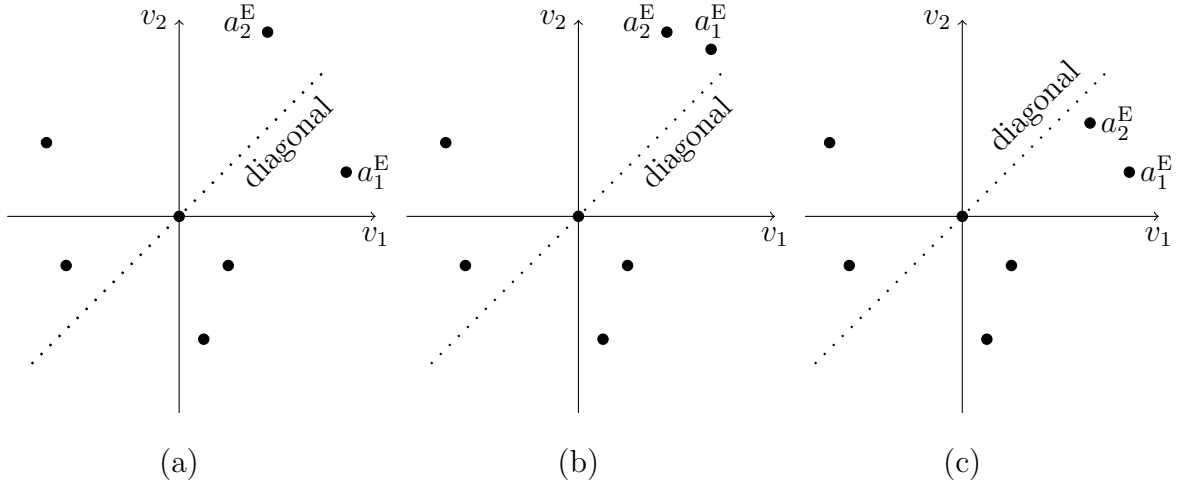


Figure 4: The three possibilities for the positions of the two efficient alternatives. (a) Both types are uncontested. (b) Type 1 is contested and type 2 is uncontested. (c) Type 1 is uncontested and type 2 is contested.

Lemma 1 (No Tradeoff Lemma) *If a type is uncontested, then in any optimal menu this type's allocation is efficient and the other type's IR constraint holds as an equality.*

Proof. Suppose that type 2 is uncontested (Assumption 1), and consider a menu $\{(a_1, p_1), (a_2, p_2)\}$ in which $a_2 \neq a_2^E$. We will show that the menu is not optimal. If both IR2 or IC2 hold as strict inequalities, then increasing p_2 slightly does not violate any of the constraints and increases the principal's profit. Suppose that IR2 or IC2 (or both) hold as equalities. Change type 2's allocation to his efficient alternative a_2^E and increase his payment by the increase $v_2(a_2^E) - v_2(a_2) > 0$ in his valuation (to leave his utility unchanged). This change clearly maintains IR1, IR2, and IC2 as in the original menu but may violate IC1. If IR2 holds as an equality in the original menu, then, to leave type 2's utility unchanged, type 2's new payment is set to $v_2(a_2^E)$. Thus, by mimicking type 2 after the change, type 1 obtains utility $v_1(a_2^E) - v_2(a_2^E)$, which is non-positive because type 2 is uncontested, so IC1 is maintained and the principal's profit strictly increases. If IC2 holds as an equality in the original menu, then, to leave type 2's utility unchanged, type 2's new payment is set to $v_2(a_2^E) - (v_2(a_1) - p_1)$. This payment (weakly) exceeds p_1 because $v_2(a_2^E) \geq v_2(a_1)$ by definition of a_2^E . So if the change violates IC1, we can further replace type 1's allocation and payment with type 2's new allocation and payment, that is, replace (a_1, p_1) with $(a_2^E, v_2(a_2^E) -$

$(v_2(a_1) - p_1))$.¹⁶ The resulting menu maintains all the constraints (because it increases type 1's utility) and strictly increases the principal's profit.

Now, consider a menu $\{(a_1, p_1), (a_2, p_2)\}$ with $a_2 = a_2^E$ and $p_1 < v_1(a_1)$, so IR1 holds as a strictly inequality. We will show that the menu is not optimal. It cannot be that both IC1 and IR2 hold as equalities, because that would imply that

$$v_1(a_1) - p_1 = v_1(a_2^E) - p_2 = v_1(a_2^E) - v_2(a_2^E) \leq 0,$$

contradicting $p_1 < v_1(a_1)$. If IC1 holds as a strict inequality, then, because IR1 holds as a strictly inequality, p_1 can be increased slightly without violating any of the constraints, which increases the principal's profit. If IR2 holds as a strict inequality, then p_1 and p_2 can both be increased by the same small amount without violating any of the constraints, which increases the principal's profit. This completes the proof of the lemma. ■

3.2 Informal Description of the Characterization

Lemma 1 implies that if both types are uncontested, then the optimal menu implements the first-best, because both types' allocations are efficient and both types obtain no information rent. If only one type is uncontested, then the first-best violates the IC constraint of the uncontested type (because the other type is contested). It thus remains to characterize the optimal menus assuming that only type 2 is uncontested (by Assumption 1). Type 1's allocation optimally resolves the tradeoff between extracting revenue from type 1 and minimizing the information of type 2. The resolution hinges on the relative probabilities of the two types. If the probability of type 1 is high, then revenue extraction is more important, so a_1 maximizes (or nearly maximizes) type 1's valuation. In other words, a_1 is efficient (or close to efficient) for type 1. This generates information rent for type 2 (because type 1 is contested).¹⁷ As the probability of type 2 increases, reducing his information rent becomes more important, even at the expense of reducing the revenue extracted from type 1. This leads to distorting a_1 away from efficiency in order to reduce type 2's information rent. If the probability of type 2 is such that it is optimal to reduce his

¹⁶With more than two types, this crucial step may fail because the resulting menu violates some type's IC constraint if the type that makes IC2 hold with equality is not the type whose IC is violated when type 2's allocation and payment are changed.

¹⁷This means that IR2 is slack so IC2 holds as an equality.

information rent to zero, so IR2 holds as an equality, it may be possible to do so without excluding type 1, unlike under increasing differences. Indeed, any alternative that type 1 values weakly more than type 2 can be allocated to type 1 at a price equal to type 1's valuation without generating information rent for type 2. These are the alternatives that lie weakly below the diagonal. Therefore, among these alternatives it is optimal to allocate to type 1 the alternative a^{BD1} he values the most.¹⁸ Thus, type 1 is excluded for a high enough probability of type 2 if and only if a^{BD1} coincides with the outside option. If IR2 holds as an equality, then both types' utilities are zero and IC2 is slack if and only if a^{BD1} lies strictly below the diagonal. This is not the first-best, however, if type 1 is not served efficiently.

For a geometric interpretation, notice that because type 1 obtains no information rent (Lemma 1), his payment is equal to his valuation for his allocated alternative a_1 . This implies that if $v_2(a_1) > v_1(a_1)$, then by mimicking type 1, type 2 can obtain utility $v_2(a_1) - v_1(a_1)$, so this is type 2's information rent. In the space of valuation pairs, the difference between the types' valuations for an alternative corresponds to the 45 degree line on which the alternative lies. A lower line corresponds to a smaller difference, or less information rent for type 2. This difference is positive if the alternative lies above the diagonal, which is the case for a_1^{E} if type 1 is contested (by definition). In this case, as the probability q of type 2 increases, it becomes optimal to allocate to type 1 an alternative with a lower valuation for type 1 that lies on a lower 45 degree line.¹⁹ For a sufficiently high probability of type 2, it may become optimal to allocate to type 1 an alternative that lies weakly below the diagonal. It is then optimal to allocate to type 1 the alternative he values the most among the ones weakly below the diagonal, that is, a^{BD1} , and to charge type 2 his valuation $v_2(a_2^{\text{E}})$. This eliminates type 2's information rent. To capture these two cases in a unified way, we add to the space of valuation pairs a "fictitious alternative" a^{FA} that corresponds to the projection of a^{BD1} on the diagonal. We then find the maximal alternative in the direction of the vector $(1, -q)$ among the ones weakly above the diagonal. If this alternative is not the fictitious alternative, then this alternative is allocated to type 1. Otherwise, alternative a^{BD1} is allocated to type 1. Figure 5 illustrates alternatives a_1^{E} , a^{BD1} , and a^{FA} , and the optimal alternative for type 1 for different values of q .

¹⁸BD1 stands for "best alternative below the diagonal for type 1."

¹⁹If an alternative with a higher valuation for type 1 lies on a lower 45 degree line, that alternative would have already been chosen because it leads to both higher revenue extraction from type 1 and lower information rent for type 2.

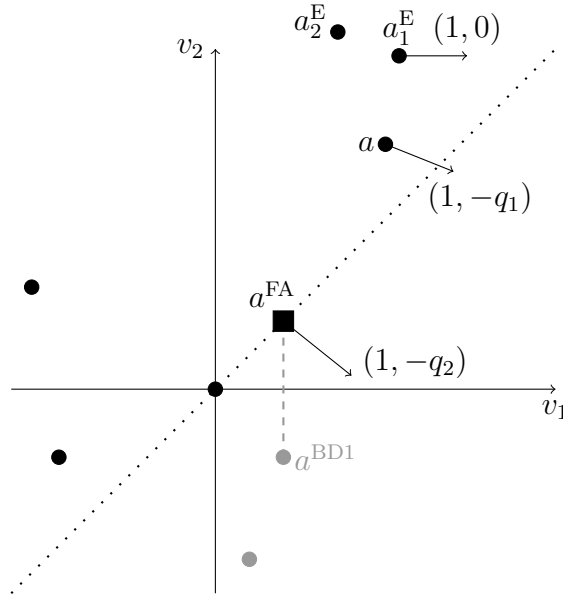


Figure 5: Alternative a_1^E is maximal in direction $(1, -q)$ when $q = 0$. Alternative a is maximal at $q = q_1$, and the fictitious alternative a^{FA} is maximal at $q = q_2$.

3.3 Formalizing the Characterization

To formalize this description and state the characterization precisely, we introduce some additional notation. Let $A^- = \{a : v_2(a) \leq v_1(a)\}$ be the set of alternatives that lie weakly below the diagonal, and let

$$a^{BD1} = \operatorname{argmax}_{a \in A^-} v_1(a) \quad (BD1)$$

be the alternative in A^- that type 1 values the most. For expositional simplicity only suppose this alternative is unique.²⁰ Now, define the “fictitious alternative” a^{FA} to satisfy

$$v_1(a^{FA}) = v_2(a^{FA}) = v_1(a^{BD1}). \quad (V)$$

If a^{BD1} lies on the diagonal, so $v_1(a^{BD1}) = v_2(a^{BD1})$, then a^{FA} coincides with a^{BD1} . Otherwise, that is, if $v_1(a^{BD1}) > v_2(a^{BD1})$, then a^{FA} is the projection of a^{BD1} on the diagonal. Alternatives a^{BD1} and a^{FA} are shown in Figure 6, (a).

²⁰Alternative a^{BD1} always exists because $v_1(0) = v_2(0) = 0$, so A^- is not empty, V is closed by assumption, and type 1’s valuation is at most $v_1(a_1^E)$.

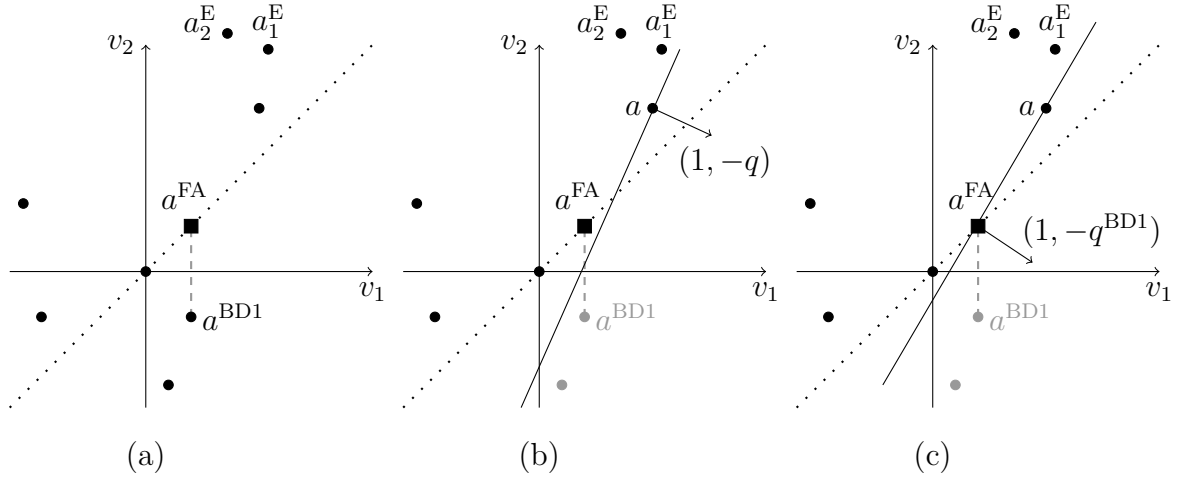


Figure 6: (a) Alternatives a^{BD1} and a^{FA} . (b) Alternative a maximizes $v_1(\cdot) - qv_2(\cdot)$ among the alternatives in $A^+ \cup \{a^{FA}\}$, and $A'_1(q) = \{a\}$. (c) q^{BD1} is the lowest q for which $a^{FA} \in A_1(q)$, because for $q < q^{BD1}$ alternative a is farther than a^{FA} in the direction of the vector $(1, -q)$.

Let $A^+ = \{a : v_1(a) \leq v_2(a)\}$ be the set of alternatives that lie weakly above the diagonal. For any $q \in [0, 1]$, let $A'_1(q)$ be the set of alternatives in $A^+ \cup \{a^{FA}\}$ that are maximal in the direction of the vector $(1, -q)$, as illustrated in Figure 6, (b),

$$A'_1(q) = \operatorname{argmax}_{a \in A^+ \cup \{a^{FA}\}} (v_1(a) - qv_2(a)).$$

Let the set $A_1(q)$ be the same as $A'_1(q)$ but with a^{FA} replaced by a^{BD1} if a^{FA} is in $A'_1(q)$. That is,

$$A_1(q) = \begin{cases} A'_1(q) & \text{if } A'_1(q) \subset A^+ \\ A'_1(q) \setminus \{a^{FA}\} \cup \{a^{BD1}\} & \text{otherwise.} \end{cases}$$

Finally, let

$$q^{BD1} = \min\{q \mid a^{BD1} \in A_1(q)\} \quad (\text{q-BD1})$$

be the lowest probability q such that $a^{BD1} \in A_1(q)$, with $q^{BD1} = 1$ if $a^{BD1} \notin A_1(q)$ for every $q < 1$.²¹ Threshold q^{BD1} is illustrated in Figure 6, (c). We now state our main

²¹Threshold q^{BD1} is well defined by continuity of $v_1 - qv_2$ in q .

result, whose proof is in Appendix A.

Theorem 1 *Suppose without loss of generality that type 2 is uncontested (Assumption 1). Then, the optimal menu implements the first-best if and only if type 1 is also uncontested. For any probability $q \in (0, 1)$ of type 2, a menu $\{(a_1, p_1), (a_2, p_2)\}$ is optimal if and only if it satisfies the following properties:*

1. *Type 2's allocation is efficient ($a_2 = a_2^E$).*
2. *Type 1's individual rationality constraint IR1 holds as an equality:*

$$p_1 = v_1(a_1).$$

3. *Type 1's alternative a_1 is any alternative in $A_1(q)$. Moreover,*

(a) *If $q < q^{\text{BD1}}$, then $A_1(q) \subset A^+ \setminus \{a^{\text{BD1}}\}$.*

(b) *If $q > q^{\text{BD1}}$, then $A_1(q) = \{a^{\text{BD1}}\}$.*

4. *If $a_1 \in A^+$, then type 2's incentive constraint IC2 holds as an equality:*

$$v_2(a_2) - p_2 = v_2(a_1) - p_1.$$

5. *If $a_1 = a^{\text{BD1}}$, then type 2's individual rationality constraint IR2 holds as an equality:*

$$p_2 = v_2(a_2^E),$$

so

$$a_1 = a^{\text{BD1}}, p_1 = v_1(a^{\text{BD1}}), a_2 = a_2^E, p_2 = v_2(a_2^E).$$

In particular, type 1's allocation a_1 pins down the menu and:

6. *The resulting profit is $v_1(a_1) - qv_2(a_1) + qv_2(a_2^E)$ if $a_1 \in A^+$ and $(1 - q)v_1(a^{\text{BD1}}) + qv_2(a_2^E)$ if $a_1 = a^{\text{BD1}}$.*
7. *Both $v_1(a_1)$ and $v_2(a_1)$ weakly decrease in q .*
8. *The utility of type 2 weakly decreases in q .*
9. *As q approaches 1, type 1 is excluded or "almost excluded," that is, $(v_1(a_1), v_2(a_2))$ approaches $(0, 0)$, if and only if a^{BD1} coincides with the outside option.*

Theorem 1 shows that the first-best is implementable if and only if both types are uncontested, in which case $a^{\text{BD1}} = a_1^{\text{E}}$. Moreover, in any optimal menu type 2 obtains his efficient alternative, IR1 holds as an equality, and type 1's alternative a_1 is maximal in the direction $(1, -q)$ among those in $A^+ \cup \{a^{\text{FA}}\}$, with a^{FA} replaced by a^{BD1} if a^{FA} is a maximizer. If $q < q^{\text{BD1}}$, so $a_1 \in A^+$, then IC2 holds as an equality. If $q > q^{\text{BD1}}$, so $a_1 = a^{\text{BD1}}$, then IR2 holds as an equality (and IC2 may be slack). If $a_1 = a^{\text{BD1}}$ and $a^{\text{BD1}} \in A^+$ (so a^{BD1} is on the diagonal), then both IC2 and IR2 hold as equalities. And if IR2 holds as an equality for some probability q of type 2, then the same is true for all higher probabilities of type 2, and the same menu $\{(a^{\text{BD1}}, v_1(a^{\text{BD1}})), (a_2^{\text{E}}, v_2(a_2^{\text{E}}))\}$ remains optimal.

With increasing differences, Theorem 1 implies that for any q and any optimal menu, type 2 is served efficiently, IR1 and IC2 hold as equalities, and the first-best is implementable if and only if it is efficient to exclude type 1 (so $a_1^{\text{E}} = 0$). This is because all the alternatives lie weakly above the diagonal, so type 2 is uncontested, and the outside option 0 is the only alternative that lies on the diagonal (see Figure 3). Thus, $a_1 \in A = A^+$ and IC2 holds as an equality in any optimal menu (by property 4 of Theorem 1), and $A^- = \{0\}$ so type 1 is uncontested if and only if $a_1^{\text{E}} = 0$. That no alternatives lie strictly below the diagonal also implies that with increasing differences random alternatives do not increase the principal's profit, as the following subsection shows.

4 Convexity and Randomization

In many applications, the set V of valuation pairs is convex. This is the case, for example, in standard price discrimination and in the vertical and horizontal product differentiation application in Section 5.1. This is also the case when random alternatives are allowed, because randomization convexifies the set of valuation pairs.

One implication of convexity is that if type 2 is uncontested (Assumption 1), then in any optimal menu type 1's alternative is in A^+ , so IC2 holds as an equality (by property 4 of Theorem 1).²² This is because if the first-best is not implementable, so type 1 is contested, then a^{BD1} lies on (and not below) the diagonal, so $a^{\text{BD1}} = a^{\text{FA}} \in A^+$ (see Figure 7). Indeed, if a^{BD1} were below the diagonal, then the line

²²Recall that with increasing differences, IC2 holds as an equality even if V is not convex. We discuss this implication of Theorem 1 at the end of Section 3.

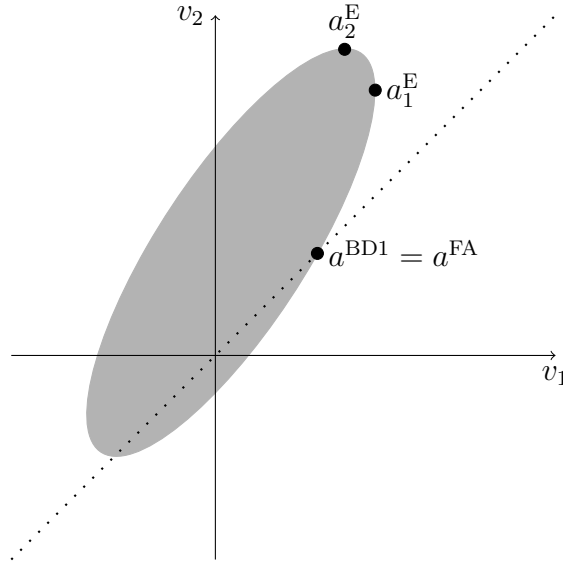


Figure 7: A convex set of valuation pairs.

that connects a^{BD1} and a_2^{E} , which is contained in V by convexity, would intersect the diagonal at a point to the right of a^{BD1} , contradicting the definition of a^{BD1} . We discuss additional implications of convexity in in Appendix E.

We also consider when allowing for random alternatives increases the principal's profit. If V for the original set of alternatives is convex, then allowing for random alternatives does not change V , so randomization does not increase the principal's profit. The following corollary characterizes when randomization increases the principal's profit if V is not convex. Its proof relies on the characterization in Appendix F of optimal menus with randomization, which follows as a corollary of Theorem 1.

Corollary 1 *Suppose that type 2 is uncontested (Assumption 1) and type 1 is contested.*

1. *If V is convex, then allowing for random alternatives has no effect; Moreover, IC2 holds as an equality for any q and in any optimal menu.*
2. *If V is not convex, then allowing for random alternatives does not increase the principal's profit for any q if and only if either (both) of the following equivalent statements hold.*

(a) $a^{\text{FA}} = a^{\text{BD1}}$ and they are both on the boundary of the convex hull of V .

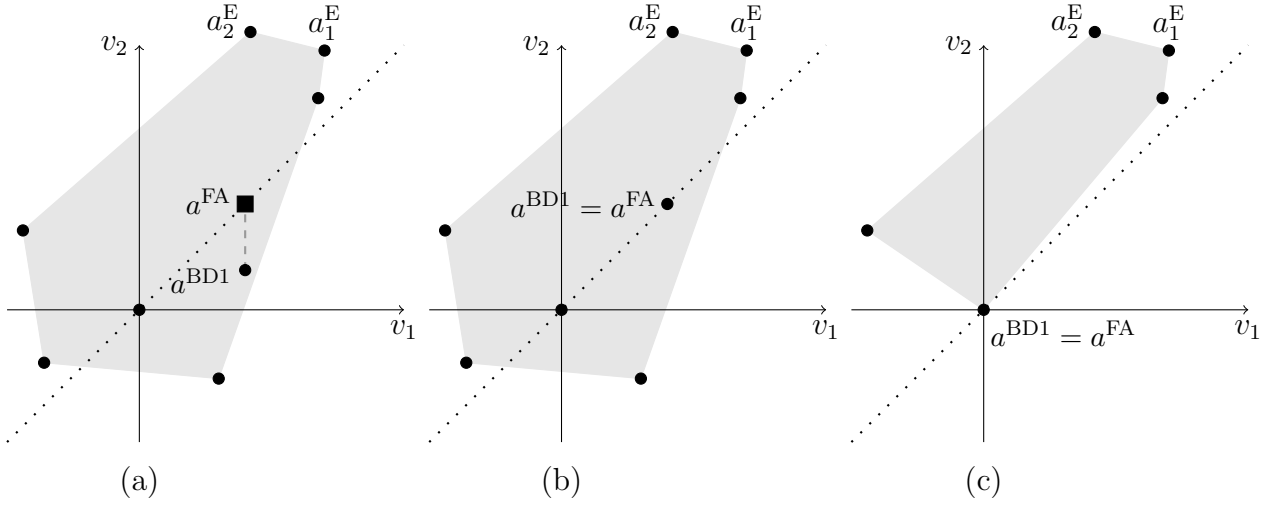


Figure 8: (a) If a^{BD1} lies strictly below the diagonal, then randomization helps. (b) Randomization helps even though a^{BD1} lies on the diagonal. (c) Randomization does not help if no alternative lies strictly below the diagonal.

(b) The highest point on the diagonal that is also in the convex hull of V is in V .

If the statements are violated, then randomization increases the principal's profit for sufficiently high $q < 1$.

Intuitively, when V is not convex randomization helps for sufficiently high q if $a^{\text{BD1}} \neq a^{\text{FA}}$ because it tightens IC2: without randomization, for sufficiently high q type 1's allocated alternative in an optimal menu is a^{BD1} , which type 1 values more than type 2. Since $a^{\text{BD1}} \neq a_1^{\text{E}}$, type 1 is not assigned his efficient alternative, so the principal would like to assign type 1 a "slightly more efficient" alternative and charge him more, which would not violate IC2 because it is slack. Randomization makes this possible, as illustrated in Figure 8, (a).

But even if $a^{\text{FA}} = a^{\text{BD1}}$, so a^{BD1} lies on the diagonal, randomization might still increase the principal's profit because $a^{\text{FA}} = a^{\text{BD1}}$ might not be on the boundary of the convex hull of V . This is illustrated in Figure 8, (b). On the other hand, Corollary 1 implies that an immediate sufficient condition for randomization *not* to increase the principal's profit is that *no* alternative lies strictly below the diagonal. This is illustrated in Figure 8, (c). This sufficient condition holds under increasing differences because then the outside option lies on the diagonal and every other alternative lies

strictly above it (see Figure 3). Thus, Corollary 1 confirms that with increasing differences, randomization does not increase the principal's profit.

5 Applications

5.1 Horizontal and Vertical Differentiation

In this application, each alternative corresponds to a product, and each product is characterized by a horizontal value and a vertical value. Both types like higher vertical values but may differ in their ideal horizontal value.

We formalize this as follows. Each alternative $a \in A$ has a “size” $s(a) \in [0, 1]$ and a “location” $\ell(a) \in [0, 1]$, with $s(0) = \ell(0) = 0$ for the outside option. We assume that all size-location pairs in $[0, 1] \times [0, 1]$ are possible, so $\{(s(a), \ell(a)) \mid a \in A\} = [0, 1] \times [0, 1]$. Each type t is also represented by a size $s(t) \in [0, 1]$ and a location $\ell(t) \in [0, 1]$. A type's size represents the intensity of his preferences, and his location represents his ideal horizontal value, so his valuation for an alternative decreases in the distance between his location and that of the alternative. Specifically, type t 's valuation for alternative a is $s(t)s(a)\cos(2\pi(\ell(t) - \ell(a)))$. If we let $\vec{u}(t)$ be a vector of size $s(t)$ and angle $2\pi\ell(t)$ and $\vec{u}(a)$ be a vector of size $s(a)$ and angle $2\pi\ell(a)$, as shown in Figure 9, (a), then type t 's valuation for alternative a is the inner product of the two vectors, $\vec{u}(t) \cdot \vec{u}(a)$. We assume that the cost of producing an alternative is linear in its size, that is, $c(a) = cs(a)$ for some constant $c \geq 0$.

The special case in which both types have the same location corresponds to standard second-degree price discrimination with quality/quantity values $s(a) \in [0, 1]$ and linear costs. This is because in this case, when considering optimal menus we can restrict attention to alternatives whose location coincides with that of the types.²³

We now describe the optimal menus for the different possible configurations of the types. If a type's size is less than c , then his efficient alternative is the outside option. If a type's size exceeds c , then his efficient alternative has size 1 and the same location as the type's. Thus, if the size of both types is less than c , then the first-best is to exclude both types, and this is implementable. Suppose that this is not the case,

²³To see this, suppose without loss of generality that both types' location is zero, that is, $\ell(t) = 0$. Any alternative a with $\ell(a) \neq 0$ can be replaced with an alternative a' that satisfies $s(a') = s(a)\cos(2\pi(\ell(a)))$ and $\ell(a') = 0$. Both types have the same valuation for alternative a' and alternative a , but the cost of alternative a' is lower.

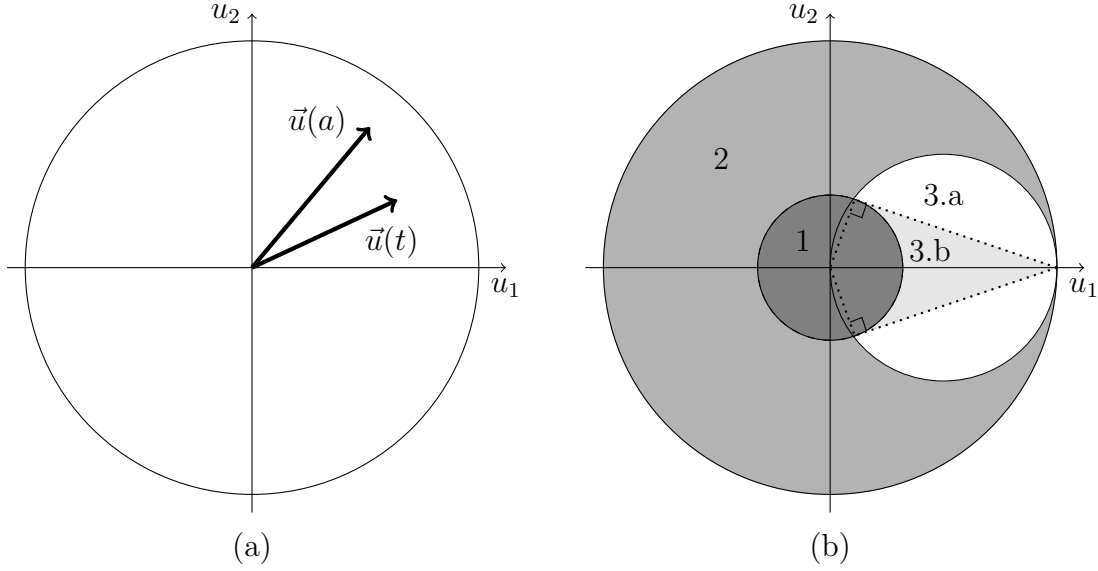


Figure 9: (a) Vector $\vec{u}(t)$ has size $s(t)$ and angle $2\pi\ell(t)$, and vector $\vec{u}(a)$ has size $s(a)$ and angle $2\pi\ell(a)$. (b) The four possible cases for type 1's location in the unit circle.

and assume without loss of generality that type 2's size weakly exceeds type 1's, so type 2 is uncontested. Moreover, normalize type 2's size to 1 and his location to 0.

Figure 9 (b) depicts the various possibilities for type 1's location in the unit circle. The first-best is implementable when type 1 is in the shaded regions 1 and 2, which is when type 1 is uncontested. In region 1, which is the circle with radius c centered at the origin, the size of type 1 is less than c , so type 1 is efficiently excluded. When type 1 lies outside of region 1, the efficient alternative of type 1 is the unit vector in his direction. In region 2, which is the area outside of region 1 that is also outside of the unshaded circle with radius $\frac{1}{2}$ centered at point $(\frac{1}{2}, 0)$, type 1 is uncontested so his optimal alternative is the efficient one (the unit vector in his direction). Intuitively, in this case type 1's ideal horizontal value (given his preference intensity) is sufficiently different from that of type 2, or, equivalently, type 1's preference intensity (given his ideal horizontal value) is sufficiently high that type 1 values his efficient alternative more than type 2. More precisely, type 1's valuation for his efficient alternative is the size of type 1, which is higher than type 2's valuation for type 1's efficient alternative, which is the projection of type 2's vector onto the unit vector in the direction of type 1, as shown in Figure 10, (a).

In regions 3.a and 3.b the first-best is not implementable, so type 1's alternative

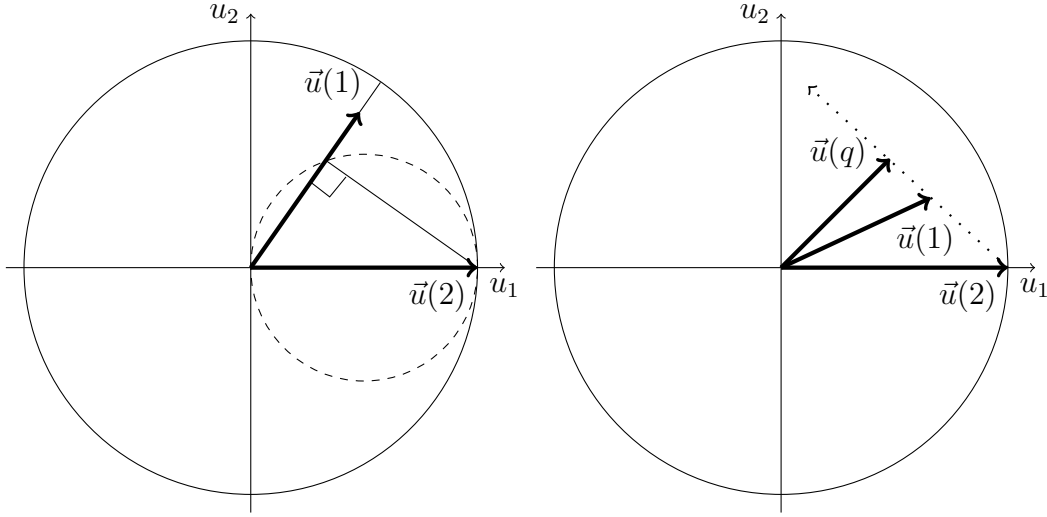


Figure 10: (a) Type 1 has a higher valuation than type 2 for the unit vector in the direction of type 1. (b) Vector $\vec{u}(q)$ is obtained by moving away from $\vec{u}(2)$ and $\vec{u}(1)$.

in any optimal menu is distorted away from efficiency. The type of distortion differs in the two regions and depends on the probabilities of the two types. To describe the distortion, for any probability q of type 2, let

$$\vec{u}(q) = \frac{1}{1-q} \vec{u}(1) - \frac{q}{1-q} \vec{u}(2)$$

be the affine combination of $\vec{u}(1)$ and $\vec{u}(2)$ with weights $\frac{1}{1-q}$ and $\frac{-q}{1-q}$. This $\vec{u}(q)$ is obtained by moving away from $\vec{u}(1)$ on the line that connects $\vec{u}(2)$ and $\vec{u}(1)$, as shown in Figure 10, (b). As q increases, $\vec{u}(q)$ moves further away from $\vec{u}(1)$ and $\vec{u}(2)$.

In region 3.a, for $q < q^{\text{BD1}}$ type 1's alternative in an optimal menu is the unit vector in the direction of $\vec{u}(q)$, and type 2 obtains information rent; for $q > q^{\text{BD1}}$ type 1's alternative is the unit vector in the direction of $\vec{u}(q^{\text{BD1}})$, and type 2 obtains no information rent. Thus, in region 3a, the distortion is in the horizontal dimension only. The threshold probability q^{BD1} is such that vector $\vec{u}(q^{\text{BD1}})$ lies on the boundary of the circle with radius $\frac{1}{2}$ centered at point $(\frac{1}{2}, 0)$, as shown in Figure 11, (a). This is because q^{BD1} is the lowest value of q such that both types have the same valuation for the unit vector in the direction of $\vec{u}(q)$, because their inner products with this unit vector coincide.²⁴ In region 3.b, for $q < q^{\text{BD1}}$ type 1's alternative is the unit vector in the direction of $\vec{u}(q)$, and type 2 obtains information rent, similarly to region 3.a,

²⁴In the space V of valuation pairs, this unit vector lies on the diagonal.

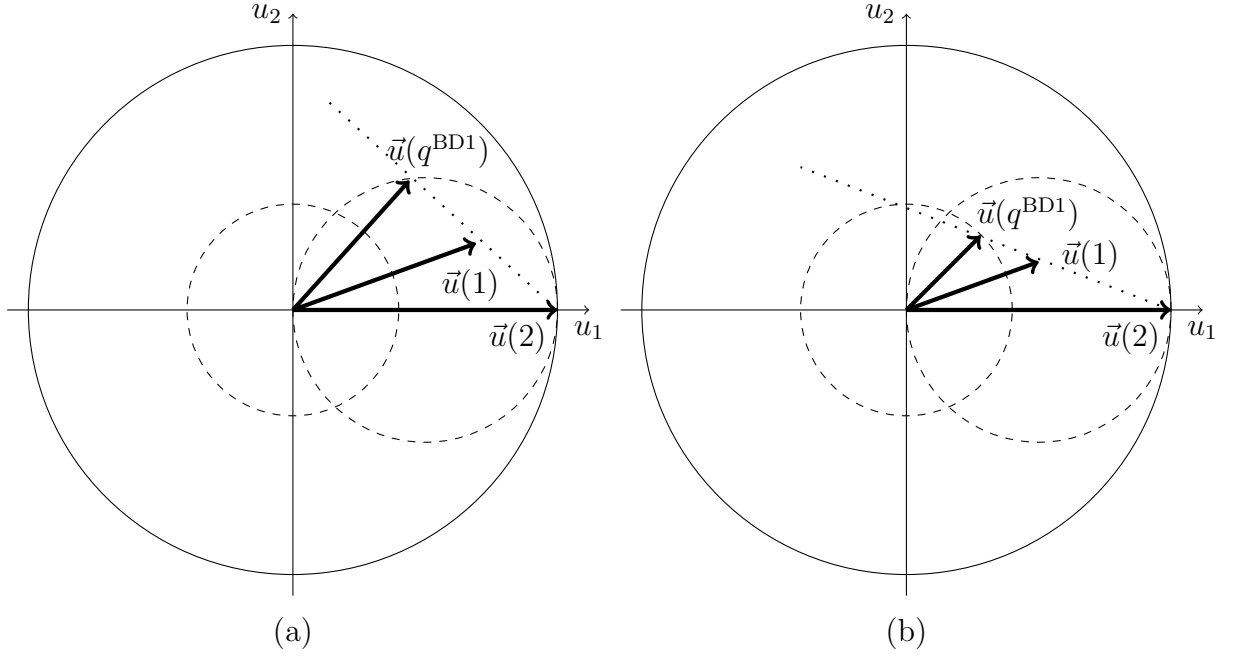


Figure 11: Threshold probability q^{BD1} in regions 3.a and 3.b.

but for $q > q^{\text{BD1}}$ type 1's alternative is the outside option and type 2 obtains no information rent. Thus, the distortion is only in the horizontal dimension for low q , but leads to inefficient exclusion of type 1 for high q . The threshold probability q^{BD1} is such that vector $\vec{u}(q^{\text{BD1}})$ lies on the boundary of the circle with radius c centered at the origin, as shown in Figure 11, (b). We formalize this discussion in Appendix H.

Notice that with no production costs ($c = 0$), type 1 can only be in region 3.b if $\vec{u}(1)$ is on the horizontal axis. Otherwise, type 1 lies above or below the line that connects $\vec{u}(2)$ to the circle with radius zero, which is the origin. So the fact that in the standard price-discrimination setting the low type might be inefficiently excluded (for sufficiently high probability of the high type) when the cost is zero is a non-generic property that only holds when both types' ideal horizontal values coincide.

5.2 Bundling with additive valuations

We consider a bundling application with a set $\{1, \dots, n\}$ of products. Each subset of products corresponds to an alternative $a \subseteq \{1, \dots, n\}$, and the empty set corresponds to the outside option $a = 0$. We allow for random alternatives, which correspond to

the set of all random subsets of products $\Delta(A)$, $A = \{a \mid a \subseteq \{1, \dots, n\}\}$.²⁵ Type t 's valuation (surplus net of production cost) for product i is $v_t(i)$, which may be negative. We assume that valuations and production costs are additive, so type t 's valuation for alternative $a \in A$ is $v_t(a) = \sum_{i \in a} v_t(i)$, with $v_t(0) = 0$. We first discuss a two-product example, and then give the characterization for any number of products.

5.2.1 An Example

Suppose that there are two products, 1 and 2, which means that there are four (deterministic) alternatives, $\emptyset$, $\{1\}$, $\{2\}$, and $\{1, 2\}$. Type 1's valuation for each product is positive, so his efficient alternative is the grand bundle $a_1^E = \{1, 2\}$. Type 2's (net) valuation is negative for product 1 and positive for product 2, so his efficient alternative is $a_2^E = \{2\}$. Even though type 2's valuation for product 1 is negative, his valuation for the grand bundle exceeds that of type 1, that is,

$$v_2(\{2\}) > v_2(\{1, 2\}) = v_2(1) + v_2(2) > v_1(1) + v_1(2) = v_1(\{1, 2\}) > v_1(\{2\}),$$

so type 2 is uncontested and type 1 is contested.²⁶ Figure 12, (a) qualitatively depicts both types' valuations for each of the two products, where type 1's valuations are on the horizontal axis and type 2's valuations are on the vertical axis.

Figure 12, (b) shows both types' valuations for all *random* alternatives. Alternatives $\{1\}$ and $\{2\}$ in panel (b) are in the same location as products 1 and 2 in panel (a). Panel (b) also depicts alternative $\{1, 2\}$ and the outside option, which corresponds to the empty set and is located at the origin. The set of all random alternatives is obtained from convexifying these four points. As shown in the figure, the fictitious alternative is an actual (random) alternative, $a^{\text{BD1}} = a^{\text{FA}}$. To distinguish the two figures, we say that panel (a) is in the “product space” and panel (b) is in the “alternative space.”

Because type 2 is uncontested and V is convex, in any optimal menu type 2 obtains his efficient alternative $a_2^E = \{2\}$ and IC2 and IR1 hold as equalities. An optimal menu is pinned down by type 1's allocation, which is any random alternative that is maximal in the direction $(1, -q)$ among those weakly above the diagonal. As shown in Figure 12, (b), the deterministic bundle $\{1, 2\}$ is maximal when $q \leq q^{\text{BD1}}$, and

²⁵Appendix K illustrates the complexities that arise when random alternatives are prohibited.

²⁶Specific valuations that satisfy all the assumptions in this section are $v_1(1) = 1, v_1(2) = 3, v_2(1) = -1$, and $v_2(2) = 6$.

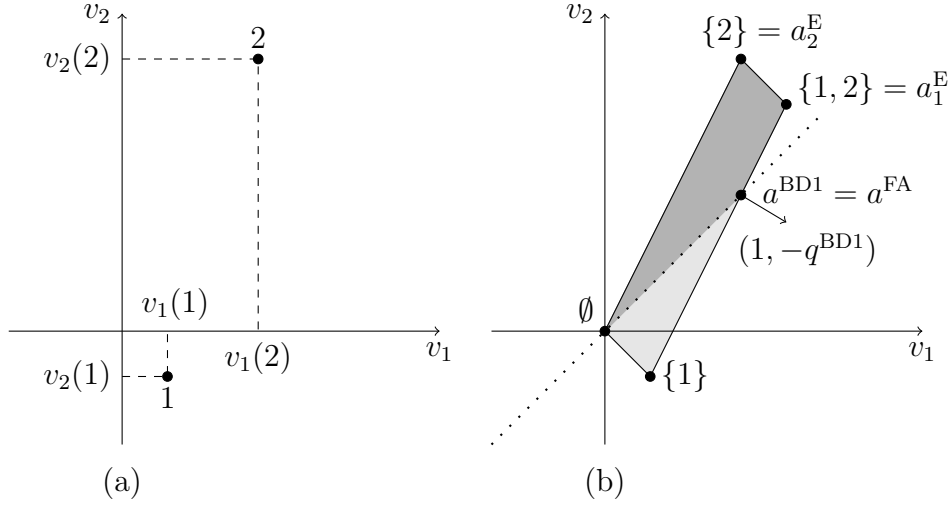


Figure 12: (a) In the product space, each point represents a product. (b) In the alternative space, each point represents a random alternative. The set A^+ of random alternatives that lie weakly above the diagonal is shaded dark.

the random alternative a^{BD1} is maximal if $q \geq q^{\text{BD1}}$. This random alternative a^{BD1} randomizes over bundles the $\{1, 2\}$ and $\{1\}$ in a way that leads to the same expected valuation for both types.

To map this solution back to the product space, consider a (deterministic) bundle that is maximal in the direction $(1, -q)$, that is, a bundle a that maximizes

$$v_1(a) - qv_2(a) = \sum_{i \in a} (v_1(i) - qv_2(i)).$$

Because the expression is additively separable across products, a maximal bundle contains a product i if $v_1(i) - qv_2(i) > 0$, does not contain the product if $v_1(i) - qv_2(i) < 0$, and may or may not contain the product if $v_1(i) - qv_2(i) = 0$.

To visualize this maximal bundle, we can draw a hyperplane in the *product space* that goes through the set of points (v_1, v_2) such that $v_1 - qv_2 = 0$, as shown in Figure 13. A product is in the maximal bundle if it is to the right of the hyperplane, is not in the bundle if it is to the left of the hyperplane, and may or may not be in the bundle if it is on the hyperplane. At the threshold q^{BD1} , the hyperplane goes through product 2, and the maximizers are either bundle $\{1, 2\}$ or bundle $\{1\}$. For $q < q^{\text{BD1}}$, as shown in Figure 13, (a), type 1's allocation is the set of products to the right of the hyperplane, namely $\{1, 2\}$. For $q \geq q^{\text{BD1}}$, as shown in Figure 13, (b),

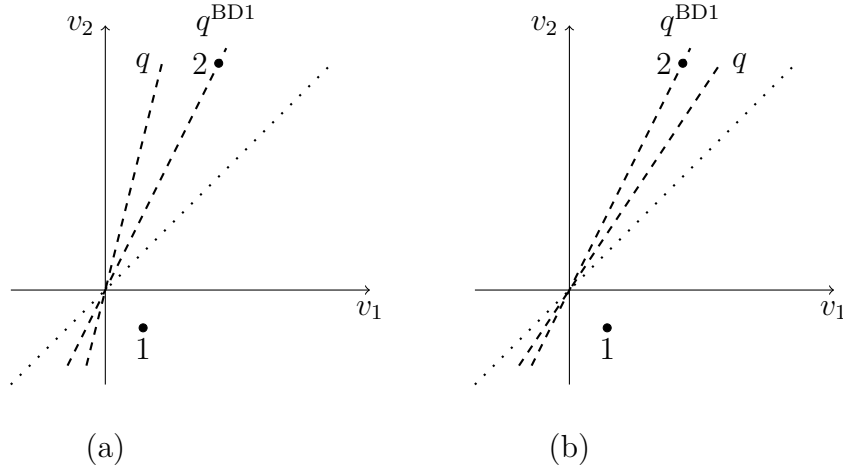


Figure 13: The hyperplane orthogonal to vector $(1, -q^{\text{BD1}})$ goes through product 2.
(a) When $q < q^{\text{BD1}}$, allocation of type 1 is products to the right of the hyperplane.
(b) When $q > q^{\text{BD1}}$, allocation of type 1 is randomized.

type 1 receives product 1 with probability 1 and product 2 with a probability that may be lower than 1.

One feature of the example is that for each product the valuation of at least one type is positive. But, perhaps surprisingly, even products for which *both* types' valuations are negative may be offered as part of an optimal menu. To see this, consider adding product 3, for which both types' valuations, $v_1(\{3\})$ and $v_2(\{3\})$, are negative. Suppose that $v_1(\{3\}) = q^3 v_2(\{3\})$ for some positive $q^3 < q^{\text{BD1}}$, as shown in Figure 14, (a). Suppose also that in the alternative space, alternative $\{1, 2, 3\}$ lies above the diagonal, as shown in Figure 14, (b).²⁷ Then, for any $q \in (q^3, q^{\text{BD1}})$, in any optimal menu type 2 obtains his efficient alternative $a_2^E = \{2\}$ and type 1 obtains the grand bundle $\{1, 2, 3\}$. Intuitively, even though allocating product 3 to type 1 reduces the surplus that can be extracted from type 1, the resulting reduction in type 2's information rent is large enough that, overall, the seller's revenue increases.²⁸

²⁷Specific valuations that satisfy all the assumptions are $v_1(1) = 1, v_1(2) = 3, v_1(3) = -1/6, v_2(1) = -1, v_2(2) = 6, \text{ and } v_2(3) = -1$.

²⁸Recall that valuations in our setting are normalized and represent willingness-to-pay net of costs. Appendix I interprets what "giving a product with a negative valuation" means in the pre-normalized setting.

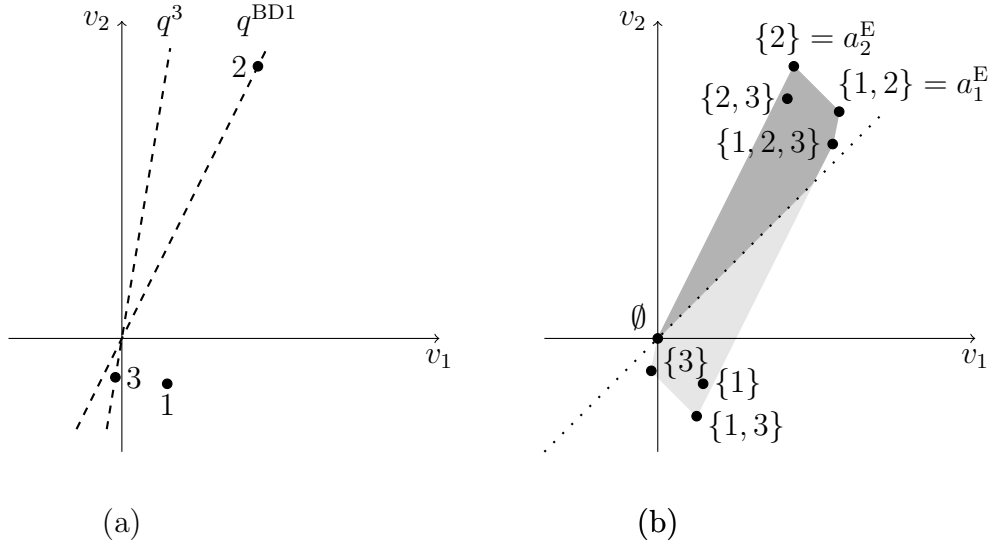


Figure 14: Representation of the 3 product example in (a) product space and (b) (random) alternative space, with the set A^+ shaded dark.

5.2.2 The Characterization with Any Number of Products

Our characterization for any number of products, which we describe informally here and formally in Appendix J, generalizes the above example. Type t 's efficient alternative $a_t^E = \{i \mid v_t(i) > 0\}$ is the set of products for which type t 's valuation is positive. Suppose without loss of generality that type 2 is uncontested, so $v_2(a_2^E) \geq v_1(a_2^E)$, and focus on the case in which type 1 is contested, so $v_2(a_1^E) > v_1(a_1^E)$ and the first-best is not implementable. In any optimal menu, type 2's allocation is a_2^E and IC2 and IR1 hold as equalities.

The optimal allocation of type 1 is described by the threshold q^{BD1} (which we will shortly identify). As shown in Figure 15, if $q < q^{\text{BD1}}$, then type 1's allocation contains every product that (in product space) is to the right of the hyperplane that goes through the set of points (v_1, v_2) such that $v_1 - qv_2 = 0$, does not contain any product to the left of that hyperplane, and may or may not contain any product that is on the hyperplane. If $q > q^{\text{BD1}}$, then what matters is the position of a product in comparison to the hyperplane corresponding to q^{BD1} , and not q : type 1's allocation contains every product that is to the right of the hyperplane that goes through the set of points (v_1, v_2) such that $v_1 - q^{\text{BD1}}v_2 = 0$, does not contain any product to the left of the hyperplane, and randomizes among the products on the hyperplane so

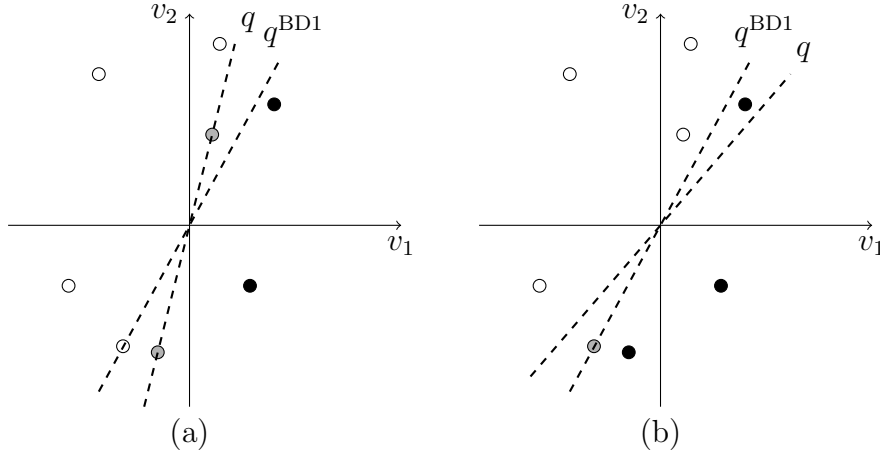


Figure 15: The optimal bundle of type 1 contains all products filled black and may or may not contain products filled gray.

that both types have the same valuation for the bundle.²⁹ Indeed, the threshold q^{BD1} is exactly where such randomization is possible: q^{BD1} is such that if we include the products on the hyperplane and in the first quadrant, then $v_2 - v_1$ is positive, and if we include the products on the hyperplane and in the third quadrant, then $v_2 - v_1$ is negative. So, by randomizing over these products appropriately, we can construct a random bundle for which $v_2 = v_1$.

This characterization clarifies which products may optimally be assigned to type 1. The products in the fourth quadrant and those in the first quadrant and below the hyperplane corresponding to q^{BD1} are always assigned to type 1. Those in the second quadrant and those in the third quadrant and above the hyperplane corresponding to q^{BD1} are never assigned to type 1. Other products might be assigned to type 1 for some values of q and not other values of q . In particular, for each product i in the first quadrant that lies above the q^{BD1} -hyperplane, define $r_i = v_1(i)/v_2(i)$. Product i is optimally assigned to type 1 whenever $q < r_i$, and is not assigned whenever $q > r_i$. Similarly, for each product i in the third quadrant that lies below the q^{BD1} -hyperplane, define $r_i = v_1(i)/v_2(i)$. Product i is not assigned to type 1 whenever $q < r_i$, and is assigned whenever $q > r_i$.

As q increases, as long as it stays below q^{BD1} , some products are added to type 1's bundle and some products are removed from it. The ones that are removed lie

²⁹If $q = q^{\text{BD1}}$, then randomization over the products on the hyperplane must be such that type 2 has a weakly higher valuation for the randomized bundle.

in the first quadrant and are above the 45 degree line through the origin. These are products for which both types have a positive valuation, but type 2 values them more, so while removing them reduces the surplus extracted from type 1, it also reduces the information rent of type 2. The products that are added are those in the third quadrant and are below the 45 degree line through the origin. These are products for which both types have a negative valuation, but type 2 values them less, so while adding them reduces the surplus extracted from type 1, it also reduces the information rent of type 2.

6 Conclusions

We characterize optimal menus in screening settings with an arbitrary set of alternatives and an agent who has quasi-linear preferences and one of two possible types. Working in the space of valuation pairs, we introduce the notion of an uncontested type, show that at least one type is uncontested, and prove that the principal faces no tradeoff between maximizing the allocation value for an uncontested type and minimizing the information rent of the other type. This implies that the first-best is implementable if and only if both types are uncontested. When only one type is uncontested, an optimal menu is characterized by the allocation of the contested type, which we identify. We provide a geometric interpretation, which also pins down which constraint(s) of the uncontested type bind.

The characterization enhances our understanding of screening with two types and the role that assumptions like increasing differences play. Some properties of optimal menus under increasing differences, such as the existence of a type that is served efficiently and a type that obtains no information rent, independently of the types' probabilities, generalize and do not require increasing differences (because at least one type is uncontested). Other properties, such as one type being indifferent between his alternative and the other type's alternative (at their respective prices) and exclusion, or "near exclusion," of one type when the probability of the other type is sufficiently high, do not always hold. Appendix D further clarifies the role of increasing differences by ordering the alternatives according to the 45 degree lines on which they lie in the space of valuation pairs. Increasing differences implies that the outside option is the unique lowest alternative in this order and no two alternatives lie on the same 45 degree line. These two properties do not hold in general, which drives the differences

between our general characterization of the optimal menus and the one in the standard setting with increasing differences.

Our characterization also clarifies how convexity of the set of valuation pairs affects the characterization of the optimal menus. This allows us to identify precisely when the ability to offer random alternatives strictly increases the principal's profit, and explains why randomization does not help the principal in the standard setting with increasing differences.

The characterization makes it possible to study canonical problems for which our current understanding is limited, such as multi-product bundling, as well as new applications, such as combining vertical and horizontal product differentiation. An intriguing direction for future research is identifying special cases in which some of the analysis applies when there are more than two types.

Appendix

A Completing the proof of Theorem 1

That the optimal menu implements the first best if and only if both types are uncontested was shown in Section 3.2. In addition, Lemma 1 proves properties 1 and 2 from Theorem 1. It remains to formalize the proof of properties 3-9 from Theorem 1, which complete the characterization of the optimal menus. As mentioned in Section 3.2, identifying type 1's allocation and payment necessitates a different approach from the one in Section 2.1 because we do not assume increasing differences. Our approach includes four steps, based on the observation that in any optimal menu either IR2 or IC2 (or both) holds as an equality.³⁰ In the first step we suppose that IR2 holds as an equality, show that this implies that IC1 holds, and solve the resulting relaxed problem.³¹ In the second step, we suppose that IC2 holds as an equality, ignore IC1, and solve the resulting relaxed problem. In the third step, we show that the solution satisfies IC1. In the fourth step, we combine the two cases (IR2 holds as an equality and IC2 holds as an equality) to obtain the characterization and prove the remaining properties from Theorem 1.

³⁰Otherwise type 2's payment can be increased without violating any of the constraints.

³¹Recall that under increasing differences IR2 can be ignored. This is not the case in general.

Step one: Suppose that IR2 holds as an equality. That $a_2 = a_2^E$ and IR1 and IR2 hold as equalities implies that $p_1 = v_1(a_1)$ and $p_2 = v_2(a_2^E)$.

Substituting these into the principal's problem and constraints, which allows us to remove IR1 and IR2, the problem becomes:

$$\begin{aligned} \max_{a_1 \in A} \quad & (1 - q)v_1(a_1) + qv_2(a_2^E) \\ \text{subject to} \quad & v_1(a_1) - v_1(a_1) \geq v_1(a_2^E) - v_2(a_2^E). \\ & v_2(a_2^E) - v_2(a_2^E) \geq v_2(a_1) - v_1(a_1). \end{aligned}$$

Rearranging, and ignoring $qv_2(a_2^E)$ in the target function, because it does not depend on a_1 , we obtain:

$$\begin{aligned} \max_{a_1 \in A} \quad & (1 - q)v_1(a_1) \\ \text{subject to} \quad & v_1(a_2^E) - v_2(a_2^E) \leq 0 \\ & v_2(a_1) - v_1(a_1) \leq 0. \end{aligned}$$

Because type 2 is uncontested, the first constraint (IC1) holds and can be ignored. We rewrite the second constraint (IC2) as $a_1 \in A^-$ (the set of alternatives that lie weakly below the diagonal) and obtain:

$$\max_{a_1 \in A^-} (1 - q)v_1(a_1). \tag{1}$$

In the solution to this problem, the principal optimally assigns to type 1 the alternative a^{BD1} , which maximizes type 1's utility among the alternatives in A^- . The resulting menu is $\{(a^{\text{BD1}}, v_1(a^{\text{BD1}})), (a_2^E, v_2(a_2^E))\}$, which generates a profit of

$$(1 - q)v_1(a^{\text{BD1}}) + qv_2(a_2^E). \tag{2}$$

Step two: Suppose that IC2 holds as an equality. That $a_2 = a_2^E$ and IR1 and IC2 hold as equalities implies that $p_1 = v_1(a_1)$ and $p_2 = v_2(a_2^E) - v_2(a_1) + v_1(a_1)$. Substituting these into the principal's problem and constraints, which allows us to

remove IC2 and IR1, the problem becomes:

$$\begin{aligned} \max_{a_1 \in A} \quad & (1 - q)v_1(a_1) + q(v_2(a_2^E) - v_2(a_1) + v_1(a_1)) \\ \text{subject to} \quad & v_2(a_2^E) - (v_2(a_2^E) - v_2(a_1) + v_1(a_1)) \geq 0, \\ & v_1(a_2^E) - (v_2(a_2^E) - v_2(a_1) + v_1(a_1)) \leq 0. \end{aligned}$$

Rearranging and ignoring $qv_2(a_2^E)$ in the target function, because it does not depend on a_1 , we obtain:

$$\begin{aligned} \max_{a_1 \in A} \quad & v_1(a_1) - qv_2(a_1) \\ \text{subject to} \quad & v_2(a_1) - v_1(a_1) \geq 0, \\ & v_2(a_1) - v_1(a_1) \leq v_2(a_2^E) - v_1(a_2^E). \end{aligned}$$

We rewrite the first constraint (IR2) as $a_1 \in A^+$ (the set of alternatives that lie weakly above the diagonal), ignore the second constraint (IC1), and obtain:

$$\max_{a_1 \in A^+} v_1(a_1) - qv_2(a_1). \quad (3)$$

This problem has the following geometric interpretation. Consider the set of valuation pairs $V^+ = \{(v_1(a), v_2(a)) : a \in A^+\}$ that corresponds to the set of alternatives A^+ . For any probability q of type 2, the principal finds a pair (v_1^q, v_2^q) in V^+ that is maximal in the direction of the vector $(1, -q)$. Any corresponding alternative $a_1 \in A^+$ generates a profit of

$$v_1(a_1) - qv_2(a_1) + qv_2(a_2^E). \quad (4)$$

Step three: We now show that IC1 holds for any solution (v_1^q, v_2^q) to the principal's problem in step 2. We showed in step 2 that IC1 is

$$v_2(a_1) - v_1(a_1) \leq v_2(a_2^E) - v_1(a_2^E),$$

so to show that IC1 holds we have to show that

$$v_2^q - v_1^q \leq v_2(a_2^E) - v_1(a_2^E).$$

For this we first prove the following lemma.

Lemma 2 For any $q' \geq q \geq 0$ and $S \subseteq \mathbb{R}^2$, if $(a^{q'}, b^{q'}) \in \max_{(a,b) \in S} (a - q'b)$ and $(a^q, b^q) \in \max_{(a,b) \in S} (a - qb)$, then $b^{q'} - a^{q'} \leq b^q - a^q$.

Proof. Suppose that $q' > q$ but $b^q - a^q < b^{q'} - a^{q'}$, that is, $b^q - b^{q'} < a^q - a^{q'}$. By definition of $(a^{q'}, b^{q'})$, we have that $a^{q'} - q'b^{q'} \geq a^q - q'b^q$, that is, $q'(b^q - b^{q'}) \geq a^q - a^{q'}$. Thus,

$$q'(b^q - b^{q'}) \geq a^q - a^{q'} > b^q - b^{q'},$$

so $b^q - b^{q'} < 0$. Consequently, because $q' > q$, we have that

$$q(b^q - b^{q'}) > a^q - a^{q'}.$$

But by definition of (a^q, b^q) , we have that $a^q - qb^q \geq a^{q'} - qb^{q'}$, that is,

$$q(b^q - b^{q'}) \leq a^q - a^{q'},$$

a contradiction. ■

By Lemma 2 applied so $S = V^+$, it is enough to show that

$$v_2^0 - v_1^0 \leq v_2(a_2^E) - v_1(a_2^E),$$

or equivalently

$$v_1(a_2^E) - v_1^0 \leq v_2(a_2^E) - v_2^0.$$

The right-hand side of the inequality is non-negative by definition of a_2^E . The left-hand side is non-positive because v_1^0 is, by definition, player 1's maximal valuation across the alternatives in A^+ , and $a_2^E \in A^+$ because type 2 is uncontested. Thus, the inequality holds.

Step four: We now combine the cases in which IR2 holds as an equality (step one) and IC2 holds as an equality (step two). Because alternative a^{BD1} maximizes type 1's valuation across the alternatives in A^- , and $v_1(a^{\text{FA}}) = v_2(a^{\text{FA}}) = v_1(a^{\text{BD1}})$ by definition of a^{FA} , the value $(1 - q)v_1(a^{\text{BD1}})$ of the solution to (1) can be written as $v_1(a^{\text{FA}}) - qv_2(a^{\text{FA}})$. Thus, the following maximization problem combines the cases:

$$\max_{a_1 \in A^+ \cup \{a^{\text{FA}}\}} v_1(a_1) - qv_2(a_1). \quad (5)$$

Denoting by $A'_1(q)$ the set of maximizers that solve (5), we have that for any maximizing alternative $a_1 \in A'_1(q)$ the corresponding profit of the principal is

$$v_1(a_1) - qv_2(a_1) + qv_2(a_2^E).$$

This is because if $a_1 \in A^+$, then in the optimum IC2 holds as an equality and the profit is given by (4), and if $a_1 = a^{\text{FA}}$, then in the optimum IR2 holds as an equality and the profit is

$$(1 - q)v_1(a^{\text{BD1}}) + qv_2(a_2^E) = v_1(a^{\text{FA}}) - qv_2(a^{\text{FA}}) + qv_2(a_2^E).$$

This proves property 6 and the first part of property 3 from Theorem 1. Property 3(a) then holds by definition of a^{BD1} .

If $a_1 \in A^+$, then a_1 solves the principal's problem, and if $a_1 = a^{\text{FA}}$, then a^{BD1} solves the principal's problem. This proves properties 4 and 5 from the statement of Theorem 1. We now prove the remaining parts of Theorem 1.

Let $\hat{V}^+ = \{(a, b) : a \leq b\} \cup \{(v_1(a^{\text{FA}}), v_2(a^{\text{FA}}))\}$ be the set of valuation pairs that corresponds to the set of alternatives $A^+ \cup \{a^{\text{FA}}\}$. For any $q \in (0, 1)$, with a slight abuse of notation denote by (v_1^q, v_2^q) a pair in $\hat{V}^+$ that corresponds to an alternative $a_1 \in A^+ \cup \{a^{\text{FA}}\}$ that solves the combined problem (5). Thus, (v_1^q, v_2^q) maximizes $a - qb$ across all valuation pairs (a, b) in $\hat{V}^+$. By Lemma 2 with $S = \hat{V}^+$, we have that $v_2^q - v_1^q$ weakly decreases in q . Now, if $a_1 \in A^+ \cup \{a^{\text{FA}}\}$ solves (5), then in the corresponding menu type 2's utility is $v_2(a_1) - v_1(a_1)$, so type 2's utility weakly decreases in q (property 8 from Theorem 1).³²

That $v_2^q - v_1^q$ weakly decreases in q also shows that if IR2 holds as an equality, so $a_1 = a^{\text{BD1}}$ is optimal for some q , then a^{BD1} is uniquely optimal for any $q' > q$ (which implies property 3(b) from Theorem 1). To see this, we first observe that since the outside option corresponds to the valuation pair $(0, 0) \in \hat{V}^+$, we must have that $v_1^q - qv_2^q \geq 0$. Together with $(v_1^q, v_2^q) \in \hat{V}^+$, we have that $v_2^q \geq v_1^q \geq qv_2^q$. This implies that $v_2^q \geq 0$, and therefore $v_1^q \geq 0$ as well. Now suppose that $q' > q$ and $a_1 = a^{\text{BD1}}$ is optimal for q . By definition of a^{BD1} and a^{FA} , $(v_1(a^{\text{FA}}), v_2(a^{\text{FA}}))$ is the highest pair in $\hat{V}^+$ on the diagonal. And since (i) a^{BD1} is optimal for q if and only

³²If $a_1 \in A^+$ then IC2 holds as an equality, so $p_2 = v_2(a_2^E) - v_2(a_1) + v_1(a_1)$ and type 2's utility is $v_2(a_1) - v_1(a_1)$, and if $a_1 = a^{\text{FA}}$, then IR2 holds as an equality and type 2's utility is $v_2(a^{\text{FA}}) - v_1(a^{\text{FA}}) = 0$.

if $(v_1^q, v_2^q) = (v_1(a^{\text{FA}}), v_2(a^{\text{FA}}))$, (ii) $(v_1^{q'}, v_2^{q'})$ lies on a weakly lower 45 degree line than (v_1^q, v_2^q) does (by Lemma 2 with $S = \hat{V}^+$), and (iii) the diagonal, on which the valuation pair $(v_1(a^{\text{FA}}), v_2(a^{\text{FA}}))$ lies, is the lowest 45 degree line in the set $\hat{V}^+$, we have that $(v_1^{q'}, v_2^{q'})$ also lies on this line. And because $q' < 1$ implies that $a - q'a < b - q'b$ for any $b > a \geq 0$, we must have $(v_1^{q'}, v_2^{q'}) = (v_1(a^{\text{FA}}), v_2(a^{\text{FA}}))$. Thus, a^{BD1} is uniquely optimal for q' .

We now show that both $v_1(a_1)$ and $v_2(a_1)$ weakly decrease in q , where a_1 is type 1's alternative in an optimal menu for q (property 7 from Theorem 1). For this we first show that both v_1^q and v_2^q weakly decrease in q . Indeed, suppose that $q' > q$ but $v_2^{q'} > v_2^q$. By definition, we have that $v_1^q - qv_2^q \geq v_1^{q'} - qv_2^{q'}$, and since $v_2^{q'} > v_2^q \geq 0$ and $q' > q$, we may conclude that

$$v_1^q - q'v_2^q > v_1^{q'} - q'v_2^{q'},$$

a contradiction. Thus, v_2^q weakly decreases in q . Now suppose that $q' > q$ but $v_1^{q'} > v_1^q$. Since $v_2^q \geq v_2^{q'}$, we have that

$$v_1^{q'} - qv_2^{q'} > v_1^q - qv_2^q,$$

a contradiction to the definition of (v_1^q, v_2^q) . That both v_1^q and v_2^q weakly decrease in q implies that both $v_1(a_1)$ and $v_2(a_1)$ weakly decrease in q because (i) if $a_1 \in A^+$ then $(v_1^q, v_2^q) = (v_1(a_1), v_2(a_1))$, (ii) if $a_1 = a^{\text{BD1}}$ then $v_1^q = v_1(a^{\text{BD1}})$ and $v_2^q \geq v_2(a^{\text{BD1}})$, and (iii) if $a_1 = a^{\text{BD1}}$ is optimal for some q , then $a_1 = a^{\text{BD1}}$ is optimal for any $q' > q$.

It remains to show property 9 from the statement of Theorem 1. Since $(0, 0) \in \hat{V}^+$, by definition of a^{BD1} we have that $v_1(a^{\text{BD1}}) \geq 0$. If $v_1(a^{\text{BD1}}) > 0$, then, since $v_1(a^{\text{FA}}) = v_1(a^{\text{BD1}})$ and v_1^q decreases in q , we have that for any q , any optimal a_1 satisfies $v_1(a_1) \geq v_1(a^{\text{BD1}}) > 0$, so type 1 is not excluded or “almost excluded.” If $v_1(a^{\text{BD1}}) = 0$, then $(0, 0)$ is the pair with the highest value in $\hat{V}^+$ on the diagonal. For $q = 1$, $(0, 0)$ maximizes (5), so for q close to 1, by continuity, the maximizer of (5) must coincide or be close to $(0, 0)$. This completes the proof of Theorem 1.

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Supplemental Appendix

B An Example

Consider a restaurant frequented by Theorists and Applied theorists, in proportions $1 - q$ and $q \in (0, 1)$, respectively. The chef knows how to prepare two dishes: steak and fish.³³ Applied theorists are willing to pay up to 60 for steak and up to 20 for fish. Theorists are willing to pay up to 30 for steak and up to $v > 0$ for fish.³⁴ Each customer can eat at most one dish. For different values of v , which dish or dishes should the restaurant offer and at what prices to maximize its profit? And how does the answer depend on the proportions of the two customer types? Theorem 1 provides a complete answer. We describe the qualitative features here, and give the details of the derivation in Section B.1.

Regardless of the proportions of the two types, it is optimal for the restaurant to offer steak and price it so that Applied theorists order it, and to charge Theorists their entire willingness to pay for the dish they order (if any). Whether and which dish the restaurant should target at Theorists depends on the values of v and q . If v is low and q is high, the restaurant should offer only steak at a price equal to Applied theorists' willingness to pay, thereby excluding Theorists. If both v and q are low, the restaurant should offer only steak at a price equal to Theorists' willingness to pay, so all customers order steak. Otherwise, the restaurant should offer both steak and fish, targeting fish at Theorists by setting its price equal to Theorists' willingness to pay for fish and adjusting the price of steak accordingly. Figure 16, (a) summarizes this by describing which dish (if any) is optimally targeted at Theorists for different values of v and q .

Now suppose that in addition to steak and fish, the restaurant can offer “surf-and-turf” dishes, which combine steak and fish in proportions that sum up to 1.³⁵ Our results also identify the optimal menus in this case, and characterize precisely

³³For simplicity only we assume that the dishes are costless to prepare.

³⁴Thus, for $v < 20$, Applied theorists are willing to pay more than Theorists for fish and also to “upgrade” from fish to steak, so Applied theorists can be thought of as “high types” and Theorists can be thought of as “low types.” But for $v > 20$, Theorists are willing to pay more than Applied theorists for fish, whereas Applied theorists are willing to pay more than Theorists customers to “upgrade” from fish to steak, so neither type is “high” or “low.”

³⁵We assume that customers' utility from a fractional steak or fish dish is linear in the fraction, so mixed dishes can be thought of as “random dishes.”

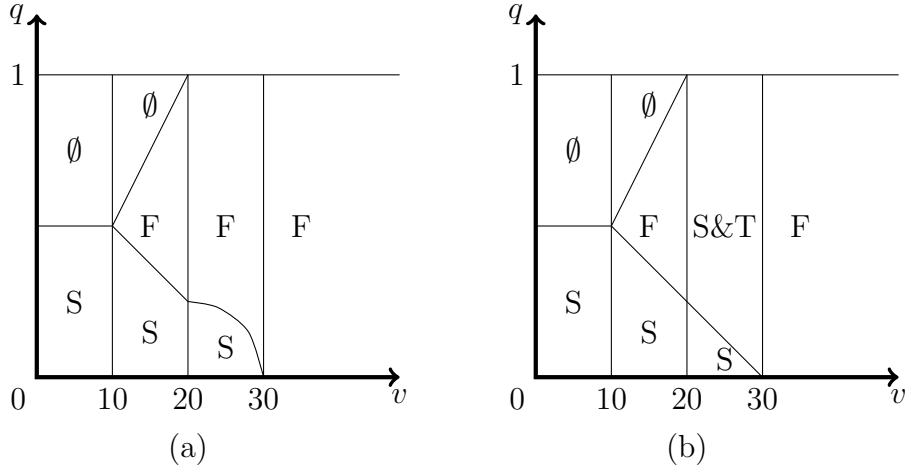


Figure 16: The dish optimally targeted at Theorists when (a) mixed dishes cannot be offered, and (b) mixed dishes can be offered. $\emptyset$ indicates exclusion, “S” indicates steak, “F” indicates Fish, and “S&T” indicates a “surf-and-turf” mixed dish.

when the restaurant strictly benefits from offering mixed dishes. With mixed dishes, it remains optimal for the restaurant to offer steak and price it so that Applied theorists order it, and to charge Theorists their entire willingness to pay for the dish they order (if any). But for intermediate values of v , there is a range of values of q (which increases in v), for which it is optimal to offer a “surf-and-turf” mixed dish targeted at Theorists. The optimal proportions of steak and fish in this dish depend only on v and do not change with q . Figure 16, (b) describes which dish (if any) is optimally targeted at Theorists for different values of v and q when mixed dishes are allowed.

B.1 Detailed of the Derivation

The efficient alternative is steak for Applied theorists. It is steak for Theorists if $v \leq 30$ and is fish if $v \geq 30$, as shown in Figure 17. Applied theorists are uncontested, so in any optimal menu their allocation is steak, and Theorists obtain no information rent. Theorists are uncontested if and only if $v \geq 30$. In this case the optimal menu implements the first-best, so steak priced at 60 and fish priced at v are offered. For the rest of the analysis suppose that $v < 30$.

We describe the optimal menus when the restaurant cannot offer mixed dishes (randomization is not allowed) and when the restaurant can offer mixed dishes. Corol-

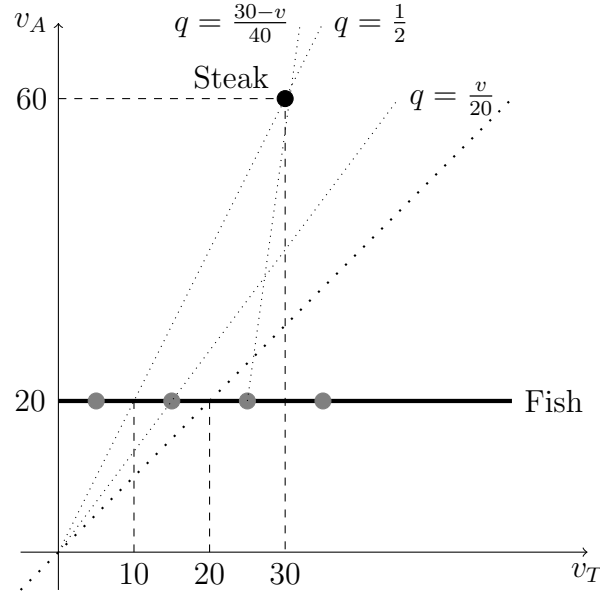


Figure 17: T stands for Theorist and A for Applied theorist. The circle shaded dark shows valuations for steak, and the circles shaded gray show some possible valuations for fish.

lary 1 shows that randomization makes a difference if and only if $20 < v < 30$ and q is sufficiently high. This is because in this case, alternative a^{BD1} , the best alternative for Theorists among those they like weakly more than Applied theorists, is fish, which lies strictly below the diagonal. When $v \leq 20$, on the other hand, a^{BD1} is the outside option, which lies on the diagonal and on the convex hull of V . Thus, in our analysis of the optimal menus below, we need to consider whether randomization is allowed only for $20 < v < 30$.

First suppose that $v \leq 10$. Then $q^{\text{BD1}} = \frac{1}{2}$ is the threshold q at which both steak and the outside option are maximal in the direction of vector $(1, -q)$. By Theorem 1, the optimal allocation for Theorists is steak if $q \leq q^{\text{BD1}} = \frac{1}{2}$, so only steak priced at 30 is offered, and is the outside option (exclusion) if $q \geq q^{\text{BD1}} = \frac{1}{2}$, so only steak priced at 60 is offered.

Now suppose that $10 \leq v \leq 20$. Then $q^{\text{BD1}} = \frac{v}{20}$ is the threshold q at which both fish and the outside option are maximal in the direction of vector $(1, -q)$. By Theorem 1, the optimal allocation for Theorists is steak if $q \leq \frac{30-v}{40}$ (which is the threshold at which both steak and fish are maximal in the direction of vector $(1, -q)$), so only steak priced at 30 is offered, is fish if $\frac{30-v}{40} \leq q \leq \frac{v}{20}$, so fish priced at v and

steak priced at $v+40$ are offered, and is exclusion if $q \geq \frac{v}{20}$, so only steak priced at 60 is offered.

Now suppose that $20 < v < 30$ and randomization is allowed. Then $q^{\text{CD}} = q^{\text{BD1}} = \frac{30-v}{40}$ is the threshold q at which both fish and steak, one below and one above the diagonal, are maximal in the direction of vector $(1, -q)$. By Corollary 3, the optimal allocation for Theorists is steak if $q \leq \frac{30-v}{40}$, so only steak priced at 30 is offered, and is “surf-and-turf” if $q \geq \frac{30-v}{40}$. The surf-and-turf lies on the diagonal, so it consists of a fraction $\frac{30}{v+10}$ of the fish dish and the remaining fraction $\frac{v-20}{v+10}$ of the steak dish. This mixed dish is priced at $\frac{60(v-10)}{v+10}$ and steak is priced 60, which extracts all surplus from both consumer types without excluding either type.

Now suppose that $20 < v < 30$ and randomization is *not* allowed. By Theorem 1, we consider the fictitious alternative a^{FA} , which in our example is an alternative for which both types have valuation v , as shown in Figure 18. Then $q^{\text{BD1}} = \frac{30-v}{60-v}$ is the threshold q at which both steak and the fictitious alternative are maximal in the direction of vector $(1, -q)$. The optimal allocation for Theorists is steak if $q \leq \frac{30-v}{60-v}$, so only steak priced at 30 is offered, and is fish if $q \geq \frac{30-v}{60-v}$, so fish priced at v and steak priced at 60 are offered, which extracts all surplus from both consumer types without excluding either type. Notice that Applied theorists strictly prefer steak priced at 60 to fish priced at v . This “slack” is what makes offering “surf-and-turf” more profitable when randomization is allowed.

C Normalizing Costs and Valuations for the Outside Option to Zero

We now show that we can normalize production costs and the valuations for the outside option to zero by changing the valuations in a way that maintains the same solutions to the principal’s problem. For this, let $\hat{v}_t(a) = v_t(a) - c(a)$ for $t = 1, 2$ and

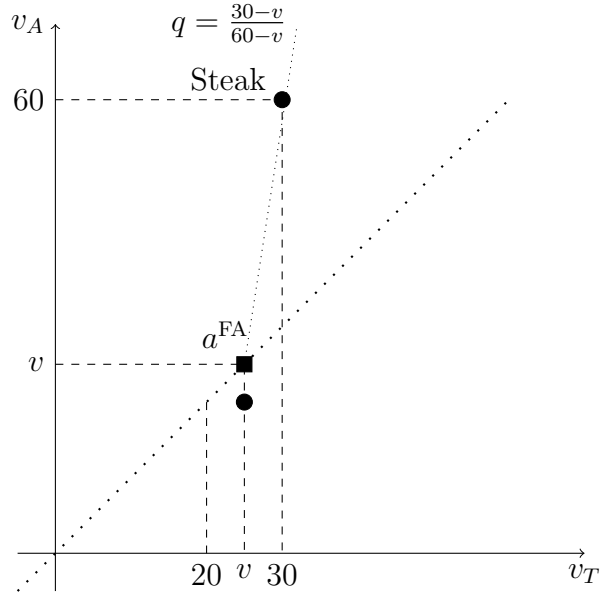


Figure 18: The case without randomization when $20 \leq v \leq 30$.

every alternative $a \in A$, and consider the following problem.

$$\begin{aligned}
& \max_{a_1, a_2 \in A \quad r_1, r_2 \in \mathbb{R}} && (1-q)r_1 + qr_2 \\
& \text{subject to} && \hat{v}_1(a_1) - r_1 \geq \hat{v}_1(0), \\
& && \hat{v}_2(a_2) - r_2 \geq \hat{v}_2(0), \\
& && \hat{v}_1(a_1) - r_1 \geq \hat{v}_1(a_2) - r_2, \\
& && \hat{v}_2(a_2) - r_2 \geq \hat{v}_2(a_1) - r_1.
\end{aligned}$$

For any a_1, a_2, r_1, r_2 , let $p_1 = r_1 + c(a_1)$ and $p_2 = r_2 + c(a_2)$. Then

$$(1-q)r_1 + qr_2 = (1-q)(p_1 - c(a_1)) + q(p_2 - c(a_2))$$

and each of IR1, IR2, IC1, and IC2 in the first problem holds if and only if the corresponding inequality in the other problem holds. Thus, the solutions to the second problem are the same as the ones to the first problem.

It remains to show that we can normalize the valuations for the outside option to zero. For this, let $\bar{v}_t(a) = \hat{v}_t(a) - \hat{v}_t(0)$ for $t = 1, 2$ and every alternative $a \in A$. Then

the second problem is equivalent to the following one.

$$\begin{aligned}
& \max_{a_1, a_2 \in A} \max_{r_1, r_2 \in \mathbb{R}} (1 - q)r_1 + qr_2 \\
& \text{subject to } \bar{v}_1(a_1) - r_1 \geq 0, \\
& \bar{v}_2(a_2) - r_2 \geq 0, \\
& \bar{v}_1(a_1) - r_1 \geq \bar{v}_1(a_2) - r_2, \\
& \bar{v}_2(a_2) - r_2 \geq \bar{v}_2(a_1) - r_1.
\end{aligned}$$

In conclusion, we can normalize production costs and the valuations for the outside option to zero by changing type t 's valuation from $v_t(a)$ to $v_t(a) - c(a) - (v_t(0) - c(0))$ for every alternative $a \in A$. The cost $c(0)$ of the outside option represents the principal's disutility from not contracting with the agent, and is typically set to 0.

D Increasing Differences

In this appendix we formally define the increasing differences assumption that is used in the screening literature to identify optimal menus and review the standard analysis under this assumption. We then discuss an alternative, weaker notion of increasing differences that our setting satisfies. This weaker notion is used in the literature to study what is *implementable* (instead of optimal).

D.1 The Standard Analysis

We consider the standard setting often studied in the screening literature and review the corresponding standard arguments used to characterize optimal menus. This setting involves alternatives that are linearly ordered, with the outside option being the lowest alternative, and assumes that one of the types, say type 2, is willing to pay more than the other type, type 1, for any increase in the alternative. Type 2 is referred to as the “high type” and type 1 is referred to as the “low type.” We formalize this standard setting as follows.

Definition 3 *Increasing differences (ID) holds if:*

1. $A \subset \mathbb{R}_+$, and

2. for any two alternatives $a, a' \in A$ with $a > a'$,

$$v_2(a) - v_2(a') > v_1(a) - v_1(a').$$

Increasing differences implies that $v_2(a) \geq v_1(a)$ for every alternative a . This property implies that IR2 can be ignored when solving for the optimal menus, which is the first step in the analysis below. Without increasing differences, a different proof is required.

Increasing differences is satisfied, for example, if $v_t(a) = t \cdot a - c(a)$ for some cost function c (recall that we normalized the valuations to be net of production costs). Increasing differences facilitates the analysis because it implies that the problem of finding an optimal menu $\{(a_1, p_1), (a_2, p_2)\}$ can be simplified as follows:³⁶

1. IR2 can be ignored because the first inequality in

$$v_2(a_2) - p_2 \geq v_2(a_1) - p_1 \geq v_1(a_1) - p_1 \geq 0,$$

follows from IC2, the second inequality follows from ID, the outside option 0 being the lowest alternative, and $v_1(0) = v_2(0) = 0$, and the last inequality follows from IR1.

2. IR1 holds as an equality (otherwise, because IR2 can be ignored, p_1 and p_2 can be increased by the same small amount).
3. IC2 holds as an equality (otherwise, because IR2 can be ignored, p_2 can be increased by a small amount).
4. IC1 can be ignored. Indeed, if it cannot be ignored, then the solution to the problem subject only to IR1 and IC2 violates IC1. And then the principal could do better by offering both types the same (alternative, price) pair: $(a_2, p_2 + \epsilon)$ for a small $\epsilon > 0$ if $p_2 \geq p_1$, and (a_1, p_1) otherwise. This trivially maintains IC2, and maintains IR1 because IC1 was violated.

Because IR1 and IC2 hold as equalities, p_1 and p_2 are pinned down by a_1 and a_2 . Thus, because IR2 and IC1 can be ignored, the principal's problem becomes:

³⁶This standard analysis mostly follows Tadelis and Segal (2005), but also highlights some features of the optimal menus. Section 2.1.3 of Bolton and Dewatripont (2004) provides a similar analysis that leads to the same conclusions under some additional assumptions.

$$\max_{a_1, a_2 \in A} (1 - q)v_1(a_1) + q(v_2(a_2) - v_2(a_1) + v_1(a_1)).$$

Grouping the elements that contain a_1 separately from the one that contains a_2 we obtain:

$$\max_{a_1, a_2 \in A} v_1(a_1) - qv_2(a_1) + qv_2(a_2).$$

Thus, we can solve for a_1 and a_2 separately. The optimal a_2 is clearly $a_2 = a_2^E$, and the optimal a_1 maximizes $v_1(\cdot) - qv_2(\cdot)$. That is, the high type is optimally served efficiently, and the low type's allocation may be distorted in order to reduce the utility $v_2(a_1) - v_1(a_1)$ (information rent) of the high type. As the probability q of the high type approaches 1, the low type's allocation a_1 becomes similar or identical to the outside option in that $(v_1(a_1), v_2(a_2))$ approaches $(0, 0)$.³⁷ These observations are summarized by the following characterization of the optimal menus.

Theorem 2 [*Tadelis and Segal (2005)*] *If increasing differences holds, then for any probability $q \in (0, 1)$ of the high type, a menu $\{(a_1, p_1), (a_2, p_2)\}$ is optimal if and only if it satisfies the following properties:*

1. *The high type's alternative is efficient ($a_2 = a_2^E$).*
2. *The high type's incentive constraint IC2 holds as an equality:*

$$v_2(a_2) - p_2 = v_2(a_1) - p_1.$$

3. *The low type's individual rationality constraint IR1 holds as an equality:*

$$p_1 = v_1(a_1).$$

4. *The low type's alternative a_1 maximizes $v_1(\cdot) - qv_2(\cdot)$.*

In particular, the low type's alternative pins down the menu and:

5. *The low type's alternative may be inefficient ($a_1 \neq a_1^E$) and he may be excluded ($a_1 = 0$).*

³⁷For $q = 1$ this is immediate because maximizing $v_1(a_1) - qv_2(a_1)$ is identical to minimizing the information rent $v_2(a_1) - v_1(a_1)$ of the high type, which is uniquely achieved in $A \subset \mathbb{R}_+$ by $a_1 = 0$. For q close to 1 this follows from a continuity argument.

6. As q approaches 1, the low type is excluded or “almost excluded,” that is, $(v_1(a_1), v_2(a_2))$ approaches $(0, 0)$.

In addition, under increasing differences the optimal menu coincides with the first-best if and only if the low type’s efficient allocation is the outside option. This is because if $a_1^E = 0$, then the menu $\{(0, 0), (a_2^E, v_2(a_2^E))\}$ satisfies all the constraints; and if $a_1^E \neq 0$, then $p_2 < v_2(a_2^E)$ in any menu $\{(a_1^E, p_1), (a_2^E, p_2)\}$ that satisfies IR1 and IC2, because $v_2(a_1^E) - v_1(a_1^E) > 0$ by increasing differences and $v_1(0) = v_2(0) = 0$. This is summarized by the following observation.

Observation 0 *If increasing differences holds, then the optimal menu implements the first-best if and only if $a_1^E = 0$.*

D.2 A Weaker Notion

Our general setting satisfies a weaker notion of increasing differences. This weaker notion is used in the literature to study which allocations are *implementable*, but not which ones are optimal. We first formalize this notion and then discuss how it differs from the standard notion of increasing differences in Definition 3 that facilitates the standard characterization in Theorem 2.

Definition 4 *Weak Increasing differences (WID) holds if there exists complete and transitive orders $\succeq_T$ and $\succeq_A$ on types and alternatives such that:*

$$t \succeq_T t', a \succeq_A a' \Rightarrow v_t(a) - v_{t'}(a) \geq v_t(a') - v_{t'}(a').$$

Write $a \succ_A a'$ if $a \succeq_A a'$ but not $a' \succeq_A a$ and $a \sim_A a'$ if $a \succeq_A a'$ and $a' \succeq_A a$.

If a setting satisfies WID, then an allocation of alternatives to types can be implemented by some IC and IR menu if and only if the allocation assigns a higher ranked alternative to a higher ranked type (see, for example, Proposition 5.5 in Börgers, 2015).

Our setting satisfies this weaker notion without imposing any additional structure on values. For this, consider ranking type 2 above type 1, $2 \succeq_T 1$, and the following order $\succeq_2$ on the set A of alternatives:

$$a \succeq_2 a' \iff v_2(a) - v_1(a) \geq v_2(a') - v_1(a').$$

That is, we assign to each alternative a the value $v_2(a) - v_1(a)$, and order the alternatives by their value. Notice, however, that WID is also satisfied with the *reverse* orders: $1 \succeq_T 2$ and the following order $\succeq_1$ on the set A of alternatives:

$$a \succeq_1 a' \iff v_1(a) - v_2(a) \geq v_1(a') - v_2(a').$$

Thus, according to WID either type can be considered the “high” or the “low” type, so WID alone cannot meaningfully single out one of the types and characterize the optimal menus. There must be some substantial difference between WID and increasing differences, which affects the optimal menus.

There are, in fact, two differences. The first is that under increasing differences, there are no ties in the rankings of alternatives, but WID allows for ties. In fact, in many applications, including the ones we consider in Section 5, it is natural for many alternatives to have the same value $v_2(\cdot) - v_1(\cdot)$. And different alternatives with the same value can play different roles in the optimal menus. For example, $a^{\text{FA}} \sim_2 0$ (and $a^{\text{BD1}} \sim_2 0$ if V is convex) but, as Theorem 1 shows, a^{FA} and a^{BD1} play a different and significant role from the outside option 0 in the optimal menus. The second difference is that increasing differences implicitly assumes that the outside option is the unique lowest alternative. This is because the outside option is 0 and property 1 in Definition 3 states that $A \subset \mathbb{R}_+$. With $\succ_2$, in contrast, the outside option may be weakly higher than any number of alternatives.

This distinction is a key difference between the case of increasing differences and the general case. Indeed, if $a_2^{\text{E}} \succeq_2 0 \succeq_2 a_1^{\text{E}}$, then both types are uncontested and the first-best is implementable. Otherwise, either $a_2^{\text{E}} \succeq_2 a_1^{\text{E}} \succ_2 0$ or $0 \succ_2 a_2^{\text{E}} \succeq_2 a_1^{\text{E}}$. In the former case, type 1 is contested (and type 2 is uncontested).³⁸ In the latter case, type 2 is contested (and type 1 is uncontested). But in both cases 0 may not be the unique $\succ_2$ -lowest alternative in A .

If $a_2^{\text{E}} \succeq_2 a_1^{\text{E}} \succ_2 0$, which is the case we have considered, but 0 is not the unique $\succ_2$ -lowest alternative in A , it might be tempting to restrict attention to the alternatives that $\succeq_2$ -exceed 0 and remove the other alternatives. This would be a mistake for two reasons. First, if $0 \succ_2 a^{\text{BD1}}$, then, as Theorem 1 shows, removing a^{BD1} affects the optimal menus when IR2 holds as an equality. This is also true if $a^{\text{BD1}} \sim_2 0$ but

³⁸This is because $a_1^{\text{E}} \succ_2 0$ means that $v_2(a_1^{\text{E}}) - v_1(a_1^{\text{E}}) > v_2(0) - v_1(0) = 0$, where the equality follows from our normalization $v_1(0) = v_2(0) = 0$.

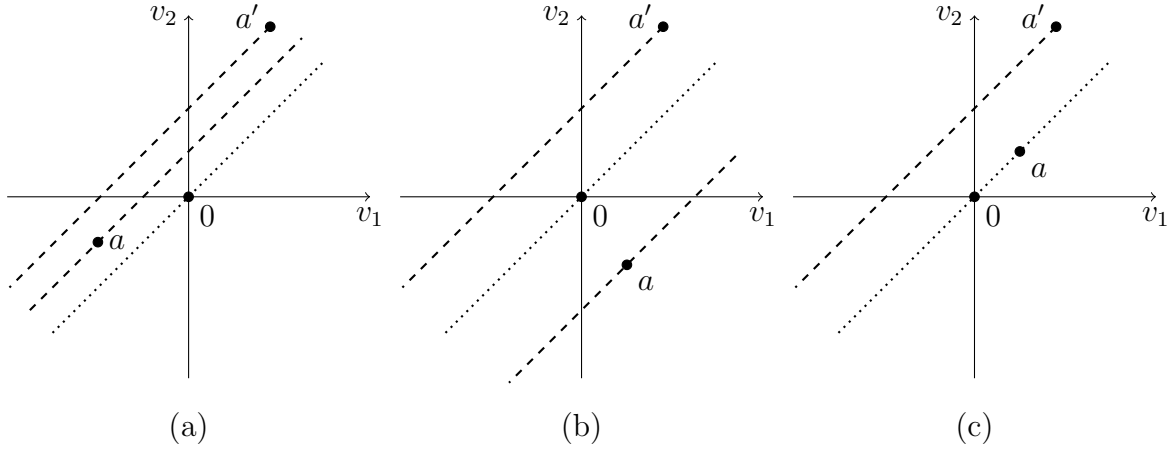


Figure 19: (a) $a' \succ_2 a \succ_2 0$. (b) $a' \succ_2 0 \succ_2 a$. (c) $a' \succ_2 0 \sim_2 a$.

$v_1(a^{\text{BD1}}) > 0$. Only if a^{BD1} coincides with the outside option is it without loss to remove the alternatives that are $\succ_2$ -lower than 0. In this case we are essentially in an increasing differences setting, IC2 always holds as an equality in an optimal menu, and type 1 is excluded or “almost excluded” as q approaches 1. Finally, if randomization is allowed and 0 is the unique $\succ_2$ -lowest alternative in A , which increasing differences satisfies, then randomization does not help.

In conclusion, what makes increasing differences special is that it identifies $\succ_2$ as the “correct” order on the alternatives and it implies that 0 is the unique $\succ_2$ -lowest alternative and that no two alternatives are $\succ_2$ -equivalent. Figure 19 illustrates the order $\succ_2$, a case in which 0 is the unique $\succ_2$ -lowest element in A in panel (a), and two cases in which 0 is not the unique $\succ_2$ -lowest element in A in panels (b) and (c).

E A Convex Set of Valuation Pairs

The set V of valuations is convex in some applications. Convexity leads to sharper predictions and a simplified formulation of Theorem 1. The reason is that with convexity, if type 1 is contested (so the first-best is not implementable) to identify type 1’s alternative a_1 in an optimal menu we can maximize $v_1(a) - qv_2(a)$ over the set of *all* alternatives A (and compare this maximum value to $v_1(a^{\text{BD1}}) - qv_2(a^{\text{BD1}})$), instead of maximizing over the constrained set $A^+ \cup \{a^{\text{FA}}\}$. This is because (i) the alternative a^{BD1} lies on (and not below) the diagonal, so $a^{\text{BD1}} = a^{\text{FA}}$, and (ii) as q increases, the first alternative $a \in A^-$ that maximizes $v_1(a) - qv_2(a)$ over the set of

all alternatives A is $a = a^{\text{BD1}}$.³⁹ This also implies that for any q , in any optimal menu IC2 holds as an equality.

To formalize this, suppose that V is convex, type 1 is contested, so the first-best is not implementable, and, without loss of generality, type 2 is uncontested (Assumption 1). We will relate type 1's alternative in an optimal menu to the set

$$B_1(q) = \operatorname{argmax}_{a \in A} v_1(a) - qv_2(a)$$

of all the alternatives that are maximal in the direction of the vector $(1, -q)$, as shown in Figure 20, (a). For this, we consider the threshold $q^{\text{OD}} \in [0, 1]$, which is the lowest value q such that some alternative in $B_1(q)$ lies on the diagonal (OD stands for “on the diagonal”).⁴⁰

Formally, let $A^\text{=}$ = $\{a : v_2(a) = v_1(a)\}$ be the set of alternatives on the diagonal, and let

$$q^{\text{OD}} = \min\{q \mid B_1(q) \cap A^\text{=} \neq \emptyset\}. \quad (\text{OD})$$

Threshold q^{OD} is shown in Figure 20, (b).

The following corollary, whose proof is in Section E.1, shows that q^{OD} is well defined and uses it to characterize the optimal menus when the set V is convex.

Corollary 2 *Suppose that type 2 is uncontested (Assumption 1), type 1 is contested, and V is convex. Then, the threshold q^{OD} defined in (OD) exists and $q^{\text{OD}} = q^{\text{BD1}}$, where q^{BD1} is defined in (q-BD1). For any probability $q \in (0, 1)$ of type 2, a menu $\{(a_1, p_1), (a_2, p_2)\}$ is optimal if and only if it satisfies the following properties:*

1. *Type 2's allocation is efficient ($a_2 = a_2^{\text{E}}$).*
2. *Type 1's individual rationality constraint IR1 holds as an equality:*

$$p_1 = v_1(a_1).$$

³⁹As we show in Appendix G, this is precisely when allowing for random alternatives does not increase the principal's profit.

⁴⁰This threshold exists because $B_1(0) = \operatorname{argmax}_{a \in A} v_1(a) = \{a_1^{\text{E}}\}$ lies weakly above the diagonal, the outside option is farther in the direction of the vector $(1, -1)$ than any alternative above the diagonal, so the alternatives in $B_1(1)$ lie weakly below the diagonal, and as q increases the alternatives in $B_1(q)$ move clockwise along the boundary of V .

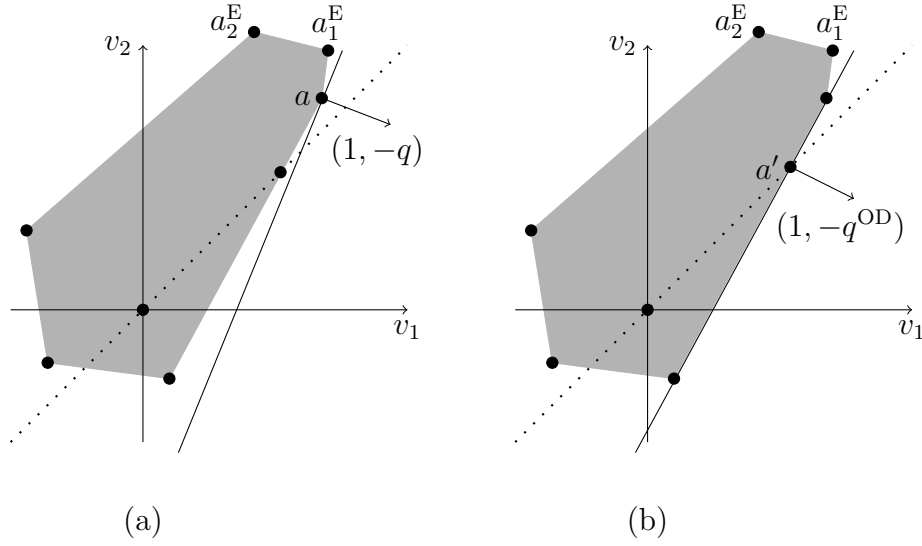


Figure 20: (a) Alternative a is maximal in the direction of the vector $(1, -q)$. (b) Alternative a' is an alternative in $B_1(q^{\text{OD}})$ that lies on the diagonal.

3. Type 1's alternative a_1 and type 2's binding constraints are specified given q as follows:

- (a) If $q < q^{\text{OD}}$, then $a_1 \in B_1(q)$ and IC2 holds as an equality.
- (b) If $q = q^{\text{OD}}$, then $a_1 \in B_1(q^{\text{OD}}) \cap A^+$ and IC2 holds as an equality. If $a_1 = a^{\text{BD1}}$ (the unique alternative in $B_1(q^{\text{OD}}) \cap A^+ \cap A^-$), then IR2 also holds as an equality.

One particular $a_1 \in B_1(q^{\text{OD}}) \cap A^+$ is the alternative in $B_1(q^{\text{OD}})$ that maximizes $v_2 - v_1$.

- (c) If $q > q^{\text{OD}}$, then $a_1 = a^{\text{BD1}}$ and both IC2 and IR2 hold as equalities.

Properties 7-9 from Theorem 1 hold as stated, and property 6 can be strengthened to the following:

6. The resulting profit is $v_1(a_1) - qv_2(a_1) + qv_2(a_2^{\text{E}})$,

and, in addition, we have the following property:

- 10. As q increases from 0 to 1, $(v_1(a_1), v_2(a_2))$ traces the boundary of V clockwise from $(v_1(a_1^{\text{E}}), v_2(a_1^{\text{E}}))$ to $(v_1(a^{\text{BD1}}), v_2(a^{\text{BD1}}))$.

E.1 Proof of Corollary 2

The following lemma, which will be used below to prove Corollary 2, shows that q^{OD} is well defined. Moreover, when $q^{\text{OD}} < 1$, we have that $B_1(q^{\text{OD}}) \cap A^=$ is the singleton $a^{\text{BD1}} = a^{\text{FA}}$, the best alternative for type 1 that lies weakly below the diagonal. In addition, q^{OD} coincides with the threshold q^{BD1} from Theorem 1.

Lemma 3 *Suppose that type 2 is uncontested (Assumption 1), type 1 is contested, and V is convex. Then*

1. *The threshold $q^{\text{OD}} \leq 1$ defined in (OD) exists.*
2. *If $q < q^{\text{OD}}$, then $B_1(q) \subseteq A^+$.*
3. *Alternative a^{BD1} lies on the diagonal, that is, $a^{\text{FA}} = a^{\text{BD1}}$.*
4. *If $q^{\text{OD}} < 1$, then $B_1(q^{\text{OD}}) \cap A^= = \{a^{\text{BD1}}\} = \{a^{\text{FA}}\}$.*
5. *Threshold q^{OD} coincides with threshold q^{BD1} defined in (q-BD1).*

Proof. We first show that a q^{OD} that satisfies (OD) exists. For this, let

$$\Delta v(q) := \min_{a \in B_1(q)} v_2(a) - v_1(a),$$

be the lowest difference between $v_2 - v_1$ among all the alternatives in $B_1(q)$, which exists because the set of valuation pairs in $B_1(q)$ is closed and convex.

We first observe that, by Lemma 2, $\Delta v(q)$ is non-increasing in q . We also observe that $\Delta v(0) > 0$ and $\Delta v(1) \leq 0$. Indeed, $B_1(0) = \{a_1^{\text{E}}\}$ and $v_2(a_1^{\text{E}}) > v_1(a_1^{\text{E}})$ because type 1 is contested, so $\Delta v(0) > 0$. And if $a \in B_1(1)$, then, by definition, for alternative $a' = 0$ we have

$$v_1(a) - v_2(a) \geq v_1(a') - v_2(a') = 0,$$

so $\Delta v(1) \leq 0$.

Now consider

$$q^* = \sup\{q \mid \Delta v(q) > 0\}.$$

Because $\Delta v(q)$ is non-increasing, we have $\Delta v(q) > 0$ for all $q < q^*$ and $\Delta v(q) \leq 0$ for all $q > q^*$.

We argue that $B_1(q^*)$ contains an alternative on the diagonal by showing that there exist alternatives $a, a' \in B_1(q^*)$ such that

$$v_2(a) - v_1(a) \geq 0 \text{ and } v_2(a') - v_1(a') \leq 0.$$

This suffices because $B_1(q^*)$ is convex. To see that such alternatives a and a' exist, consider a sequence $a(q)$ of alternatives as q approaches q^* from below, where $a(q)$ is an alternative in $B_1(q)$ that minimizes $v_2 - v_1$. Because V is closed, there exists some $v^* \in V$ such that

$$(v_1^*, v_2^*) = \lim_{q \uparrow q^*} (v_1(a(q)), v_2(a(q))).$$

Let a^* be an alternative with $v^* = (v_1(a^*), v_2(a^*))$. Because $v_1(a) - qv_2(a)$ is continuous in q , we have that $a^* \in B_1(q^*)$. And because $v_2(a(q)) - v_1(a(q)) > 0$ for all q in the sequence,

$$v_2(a^*) - v_1(a^*) \geq 0.$$

A similar argument, but approaching q^* from above, shows that there exists an alternative $a' \in B_1(q^*)$ such that

$$v_2(a') - v_1(a') \leq 0.$$

We now claim that $q^{\text{OD}} = q^*$. We have already shown that $B_1(q^*)$ contains an alternative on the diagonal. That there is no $q < q^*$ with this property follows because $\Delta v(q) > 0$ for all $q < q^*$ means that $v_2(a) - v_1(a) > 0$ for all $a \in B_1(q)$ and $q < q^*$.

We now show the second statement. As argued above, for any $q < q^{\text{OD}}$, $a \in B_1(q)$, and $a' \in B_1(q^{\text{OD}})$, we have that $v_2(a) - v_1(a) \geq v_2(a') - v_1(a')$. So, because $v_2(a') - v_1(a') = 0$ for some $a' \in B_1(q^{\text{OD}})$, we have that $v_2(a) - v_1(a) \geq 0$ for all $a \in B_1(q)$, so $B_1(q) \subseteq A^+$.

To see that a^{BD1} lies on the diagonal, so $a^{\text{FA}} = a^{\text{BD1}}$, suppose that a^{BD1} lies strictly below the diagonal, so $v_1(a^{\text{BD1}}) > v_2(a^{\text{BD1}})$. Since type 1 is contested, a_1^{E} lies strictly above the diagonal. Thus, the line connecting a^{BD1} and a_1^{E} intersects the diagonal

at a single point, which corresponds to some alternative a by convexity of V . And $v_1(a) > v_1(a^{\text{BD1}})$ because $v_1(a_1^{\text{E}}) > v_1(a^{\text{BD1}})$ by definition of $v_1(a_1^{\text{E}})$. This contradicts the definition of a^{BD1} , so we conclude that a^{BD1} lies on the diagonal.

Now suppose that $q^{\text{OD}} < 1$. The set $B_1(q^{\text{OD}})$ is the set of alternatives in V that lie on some line L whose slope strictly exceeds 1 because L is orthogonal to vector $(1, -q^{\text{OD}})$. Because V is convex and closed, $B_1(q^{\text{OD}})$ is a convex and closed segment of L , which by definition of q^{OD} has non-empty intersection with the diagonal. Because the slope of this segment strictly exceeds 1, the point of intersection is unique. Let $\tilde{a}_1$ be the unique element in $B_1(q^{\text{OD}}) \cap A^=$.

To see that $\tilde{a}_1 = a^{\text{BD1}}$, consider any alternative $a \in A^-$, so $v_1(a) \geq v_2(a)$, and write

$$(1 - q^{\text{OD}})v_1(a) \leq v_1(a) - q^{\text{OD}}v_2(a) \leq v_1(\tilde{a}_1) - q^{\text{OD}}v_2(\tilde{a}_1) = (1 - q^{\text{OD}})v_1(\tilde{a}_1),$$

where the second inequality follows because $\tilde{a}_1 \in B_1(q^{\text{OD}})$, and the equality follows because $\tilde{a}_1$ is on the diagonal. That $(1 - q^{\text{OD}})v_1(a) \leq (1 - q^{\text{OD}})v_1(\tilde{a}_1)$ implies that $v_1(a) \leq v_1(\tilde{a}_1)$, so $\tilde{a}_1$ coincides with a^{BD1} by definition of a^{BD1} (which is assumed to be unique).

We conclude the proof by showing that $q^{\text{OD}} = q^{\text{BD1}}$, where q^{BD1} is defined in (q-BD1). We showed in the first part of the proof that $\Delta v(q) > 0$ for every $q < q^{\text{OD}}$. This implies, by definition of q^{BD1} , that $q^{\text{OD}} \leq q^{\text{BD1}}$. In particular, if $q^{\text{OD}} = 1$, then $q^{\text{OD}} = q^{\text{BD1}}$. And we saw that if $q^{\text{OD}} < 1$, then $a^{\text{BD1}} \in B_1(q^{\text{OD}})$. This implies, again by definition of q^{BD1} , that $q^{\text{BD1}} \leq q^{\text{OD}}$. ■

Given Lemma 3, we now complete the proof of Corollary 2.

Proof of Corollary 2. Property 6 follows from property 6 of Theorem 1 because IC2 holds as an equality for all $q \in (0, 1)$. Property 10 follows from properties 7 and 8 of Theorem 1 along with the fact that, by linearity, the alternatives in $A_1(q)$ lie on the boundary of V .

We now prove properties 3(a), (b), and (c). Because $a^{\text{FA}} = a^{\text{BD1}} \in A^=$ (by statement (3) of Lemma 3),

$$\operatorname{argmax}_{a \in A^+ \cup \{a^{\text{FA}}\}} v_1(a) - qv_2(a) = \operatorname{argmax}_{a \in A^+} v_1(a) - qv_2(a),$$

so a_1 is any alternative in

$$\operatorname{argmax}_{a \in A^+} v_1(a) - qv_2(a).$$

Consider $q < q^{\text{OD}}$. By statement (2) of Lemma 3, we have that $B_1(q) \subseteq A^+$, that is,

$$\operatorname{argmax}_{a \in A} v_1(a) - qv_2(a) = \operatorname{argmax}_{a \in A^+} v_1(a) - qv_2(a),$$

so a_1 is any alternative in $B_1(q)$, and IC2 holds as an equality by property 4 of Theorem 1.

Consider $q = q^{\text{OD}}$. Because $B_1(q^{\text{OD}})$ has an alternative in A^+ ,

$$\max_{a \in A} v_1(a) - qv_2(a) = \max_{a \in A^+} v_1(a) - qv_2(a),$$

so a_1 is any alternative in $B_1(q^{\text{OD}}) \cap A^+$. The binding constraints follow from properties 4 and 5 of Theorem 1.

Consider $q > q^{\text{OD}}$. Since $q^{\text{OD}} = q^{\text{BD1}}$ and $a^{\text{BD1}} \in A^= \subset A^+$ (by statement (3) of Lemma 3), property 3(b) of Theorem 1 implies that $a_1 = a^{\text{BD1}}$, and properties 4 and 5 of Theorem 1 imply that both IC2 and IR2 hold as equalities. ■

F Allowing for Randomization

We use Corollary 2 to characterize, for a given set of alternatives A , optimal menus that can include random alternatives. A random alternative is an element of $\Delta(A)$, and type t 's valuation for random alternative x is $v_t(x) = E_{a \sim x}[v_t(a)]$.⁴¹ We assume that type 2 is uncontested (Assumption 1), as we did in Theorem 1, and that type 1 is contested (otherwise the first-best is implementable and randomization has no effect on the optimal menus).

We begin with a generalization of the definition of the threshold q^{OD} from (OD)

⁴¹The principal's cost of producing x is $c(x) = E_{a \sim x}[c(a)]$, which is equal to 0 because we normalized the production costs of all alternatives to 0.

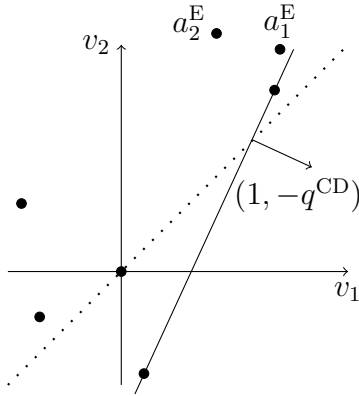


Figure 21: There is no q such that $B_1(q)$ contains an alternative on the diagonal, but $B_1(q^{\text{CD}})$ contains an alternative above and an alternative below the diagonal.

that does not require V to be convex. Recall from Appendix E that

$$B_1(q) = \operatorname{argmax}_{a \in A} v_1(a) - qv_2(a)$$

is the set of alternatives that are maximal in the direction of the vector $(1, -q)$, and for a convex V , q^{OD} is the lowest threshold at which an alternative in $B_1(q^{\text{OD}})$ lies on the diagonal. With a non-convex V , there may not be a maximizer on the diagonal for any q , because the alternatives in $B_1(q)$ might “jump” at some point from above the diagonal to below it, as Figure 21 shows. Nevertheless, there exists a threshold at which $B_1(q)$ “crosses” the diagonal, that is, some alternative in $B_1(q)$ lies weakly above the diagonal and some alternative in $B_1(q)$ lies weakly below the diagonal. Formally, let

$$q^{\text{CD}} = \min\{q \mid B_1(q) \cap A^+ \neq \emptyset \text{ and } B_1(q) \cap A^- \neq \emptyset\}. \quad (\text{CD})$$

This threshold q^{CD} is shown in Figure 21 (CD stands for “crosses the diagonal”) and coincides with the threshold q^{BD1} from Theorem 1 when q^{BD1} is computed with respect to the set of random alternatives. The following corollary characterizes the optimal menus when randomization is allowed.

Corollary 3 *Suppose that type 2 is uncontested (Assumption 1), type 1 is contested, and randomization is allowed. Then, the threshold q^{CD} defined in (CD) exists and $q^{\text{CD}} = q^{\text{BD1}}$, where q^{BD1} , defined in (q-BD1), is computed with respect to the set of ran-*

dom alternatives. For any probability $q \in (0, 1)$ of type 2, a menu $\{(x_1, p_1), (x_2, p_2)\}$ of random alternatives is optimal if and only if it satisfies the following properties:

1. Type 2's allocation is efficient ($x_2 = a_2^E$).
2. Type 1's individual rationality constraint IR1 holds as an equality:

$$p_1 = v_1(x_1).$$

3. Type 1's random alternative x_1 and type 2's binding constraints are specified given q as follows:

- (a) If $q < q^{CD}$, then x_1 is any distribution over the alternatives in $B_1(q)$, and IC2 holds as an equality.
- (b) If $q = q^{CD}$, then x_1 is any distribution over the alternatives in $B_1(q^{CD})$ such that

$$v_2(x_1) \geq v_1(x_1),$$

and IC2 holds as an equality. If the above inequality holds as an equality, then IR2 also holds as an equality.

One particular x_1 that satisfies the above inequality assigns probability 1 to the alternative a in $B_1(q^{CD})$ that maximizes $v_2 - v_1$.

- (c) If $q > q^{CD}$, then x_1 is any distribution over the alternatives in $B_1(q^{CD})$ such that

$$v_2(x_1) = v_1(x_1),$$

and both IC2 and IR2 hold as equalities.

One particular x_1 that satisfies the above equality randomizes only over the two alternatives in $B_1(q^{CD})$ that maximize and minimize $v_2 - v_1$ (if the two alternatives coincide, then x_1 assigns probability 1 to this alternative).

Proof. Denote by

$$V^R = \{(v_1(x), v_2(x)) : x \text{ is a random alternative}\}$$

the set of valuation pairs associated with random alternatives. Because V^R is convex, Corollary 2 characterizes the optimal menus. Thus, properties 1 and 2 hold, and it remains only to prove property 3.

For this consider the set

$$B^R(q) = \operatorname{argmax}_{x \in \Delta(A)} v_1(x) - qv_2(x),$$

of maximizers of $v_1 - qv_2$ over all random alternatives. By linearity of $v_1 - qv_2$, $B^R(q)$ is the convex hull of the set $B_1(q)$ of all (non-random) alternatives that maximize $v_1 - qv_2$, that is, $\Delta(B_1(q)) = B^R(q)$, so $B_1(q) \subseteq B^R(q)$. Because V^R is convex, Lemma 3 shows that there is a lowest threshold, call it q^{OD-R} , such that some (random) alternative $x \in B^R(q^{OD-R})$ is on the diagonal, that is,

$$v_1(x) = v_2(x).$$

Notice that for any q , because $\Delta(B_1(q)) = B^R(q)$, there is a random alternative $x \in B^R(q)$ with $v_1(x) - v_2(x) = 0$ if and only if there are two (deterministic) alternatives $a, a' \in B_1(q)$ such that $v_1(a) - v_2(a) \geq 0$ and $v_1(a') - v_2(a') \leq 0$, that is, $B_1(q) \cap A^+$ and $B_1(q) \cap A^-$ are non-empty. Thus, $q^{CD} = q^{OD-R} = q^{BD1}$, where the second equality follows from part 5 of Lemma 3.

Now consider the three cases in property 3 and apply Corollary 2. If $q < q^{CD} = q^{OD-R}$, then x_1 is any alternative in $B^R(q) = \Delta(B_1(q))$, that is, any distribution over the alternatives in $B_1(q)$, and IC2 holds as an equality.

Next consider $q = q^{CD} = q^{OD-R}$. Then x_1 is any random alternative in $B^R(q^{OD-R})$ that is on or above the diagonal. These are distributions x over alternatives in $B_1(q^{CD})$ such that

$$v_2(x) \geq v_1(x),$$

and IC2 holds as an equality. If the inequality holds as an equality, then IR2 also holds as an equality.

Finally consider $q > q^{CD} = q^{OD-R}$. Then, x_1 is the “unique alternative” in $B_1(q^{OD-R}) \cap A^+ \cap A^-$. Thus, x_1 is any random alternative in $B^R(q^{OD-R})$ that is on the diagonal, that is, any distribution x over the alternatives in $B_1(q^{CD})$ such that $v_1(x) = v_2(x)$, and both IC2 and IR2 hold as equalities. ■

G Proof of Corollary 1

Proof. We use Corollary 3 to identify the set of optimal menus when randomization is allowed. Randomization increases the principal's profit if and only if there is no deterministic menu in that set.

Because type 2 is uncontested, its allocation is efficient and therefore deterministic. Randomization can therefore increase the principal's profit only through type 1's allocation. If $q < q^{\text{BD1}}$, where q^{BD1} , defined in (q-BD1), is computed with respect to the set of random alternatives, then by part 3.(a) of Corollary 3, type 1's random alternative is any distribution over the set $B_1(q)$ of deterministic alternatives, so randomization does not help. If $q = q^{\text{BD1}}$, by part 3.(b) of Corollary 3, one optimal allocation for type 1 assigns probability 1 to the alternative in $B_1(q^{\text{BD1}})$ that maximizes $v_2 - v_1$, so randomization does not help.

Now suppose $q > q^{\text{BD1}}$. Then, by part 3.(c) of Corollary 3, type 1's optimal allocation is any distribution x_1 over the alternatives in $B_1(q^{\text{BD1}})$ such that $v_2(x_1) = v_1(x_1)$. That is, the optimal allocation of type 1 is at the intersection of the diagonal and the convex hull of V . So randomization does not help if and only if there is a (deterministic) alternative at that intersection. This holds if and only if (1) $a^{\text{FA}} = a^{\text{BD1}}$ (for the original set of deterministic alternatives) and (2) both of these two alternatives are on the boundary of the convex hull of V . ■

H Characterizing the Optimal Menus in the Horizontal and Vertical Differentiation Application

Consistent with our model, we normalize costs to zero by reflecting the cost in the valuations and writing $v_t(a) = s(t)s(a)\cos(2\pi(\ell(t) - \ell(a))) - cs(a)$. To simplify notation we write $\cos(t, a)$ instead of $\cos(2\pi(\ell(t) - \ell(a)))$, so we have $v_t(a) = s(t)s(a)\cos(t, a) - cs(a)$. In inner product form, this is

$$v_t(a) = \vec{u}(t) \cdot \vec{u}(a) - cs(a).$$

Without loss of generality we assume that the size of type 2 weakly exceeds that of type 1, and we normalize the size of type 2 to 1, so $s(2) = 1$. We also assume without loss of generality that the location of type 2 is 0.

The following result formalizes the description of the optimal menus from Section 5.1.

Proposition 1 *Suppose without loss of generality that $s(2) = 1 \geq s(1)$, $l(2) = 0$, and $c < 1$. For any probability $q \in (0, 1)$ of type 2, a menu $\{(a_1, p_1), (a_2, p_2)\}$ is optimal if and only if it satisfies the following properties:*

1. *Type 2's allocation is efficient ($a_2 = a_2^E$ is the unit vector $(0, 1)$).*
2. *Type 1's individual rationality constraint IR1 holds as an equality:*

$$p_1 = v_1(a_1).$$

3. *If type 1 is in region 1 in Figure 9, (b), then the first-best is implementable and a_1 is the outside option.*
4. *If type 1 is in region 2 in Figure 9, (b), then the first-best is implementable and a_1 is the unit vector in the direction of $\vec{u}(1)$.*
5. *If type 1 is in region 3.a or 3.b in Figure 9, (b), then a_1 and type 2's binding constraints are specified given q as follows.*

(a) *If type 1 is in region 3.a, then q^{OD} is such that $\vec{u}(q^{\text{OD}})$ lies on the boundary of the circle with radius $\frac{1}{2}$ centered at $(\frac{1}{2}, 0)$ and:*

- i. *If $q \leq q^{\text{OD}}$, then a_1 is the unit vector in the direction of $\vec{u}(q)$ and IC2 holds as an equality.*
- ii. *If $q > q^{\text{OD}}$, then a_1 is the unit vector in the direction of $\vec{u}(q^{\text{OD}})$ and both IC2 and IR2 hold as equalities.*

(b) *If type 1 is in region 3.b, then q^{OD} is such that $\vec{u}(q^{\text{OD}})$ lies on the boundary of the circle with radius c centered at $(0, 0)$ and:*

- i. *If $q < q^{\text{OD}}$, then a_1 is the unit vector in the direction of $\vec{u}(q)$ and IC2 holds as an equality.*
- ii. *If $q = q^{\text{OD}}$, then a_1 is either the outside option or any vector in the direction of $\vec{u}(q)$, and both IC2 and IR2 hold as equalities.*
- iii. *If $q > q^{\text{OD}}$, then a_1 is the outside option and both IC2 and IR2 hold as equalities.*

As outlined in Section 5.1, the efficient alternative of each type is the unit vector in the direction of that type if the size of the type is more than c , and is the outside option if the size is less than c . Moreover, the first-best is clearly implementable if type 1 is either in region 1 or region 2 of Figure 9 (b). To complete the proof of Proposition 1, it thus remains to prove property 5, which corresponds to type 1 being in region 3.a or 3.b.

To characterize the optimal menus in this case, we observe that the set of valuation pairs of the two types across all alternatives is convex. Thus, we can apply Corollary 2, which shows that type 2's allocation is efficient and IR1 and IC2 hold as equalities. Any optimal menu is therefore pinned down by type 1's alternative a_1 . To identify a_1 , we need to identify the set $B_1(q)$ of alternatives that maximize $v_1(a) - qv_2(a)$ across all alternatives, where q is the probability of type 2.

Among all the alternatives that have some fixed size $\bar{s}$, the alternative a that maximizes the objective is in the direction of $\vec{u}(q)$, in which case the value of the objective is $\vec{u}(q) \cdot \vec{u}(a) - c\bar{s} = (s(\vec{u}(q)) - c)\bar{s}$. Thus, we can focus on maximizing $(s(\vec{u}(q)) - c)s(a)$. If $s(\vec{u}(q)) > c$, then the unique maximizer is the unit vector in the direction of $\vec{u}(q)$. If $s(\vec{u}(q)) < c$, then the unique maximizer is the outside option. If $s(\vec{u}(q)) = c$, then the outside option and every vector in the direction of $\vec{u}(q)$ are maximizers. This characterizes the set $B_1(q)$.

Now consider two cases. The first case is when $\vec{u}(1)$ lies in region 3.a. In this case, the line that connects $\vec{u}(2)$ and $\vec{u}(1)$ never enters the circle with radius c , as shown in Figure 22. So $\vec{u}(q)$ is outside the circle with radius c , that is, $s(\vec{u}(q)) > c$, for all q and therefore $B_1(q)$ consists of the unit vector in the direction of $\vec{u}(q)$, which we denote by $a(q)$. The threshold q^{OD} defined in (OD) is such that

$$v_1(a(q^{\text{OD}})) = v_2(a(q^{\text{OD}})).$$

For both types to have the same valuation for $a(q^{\text{OD}})$, the projections of $\vec{u}(1)$ and $\vec{u}(2)$ onto $a(q^{\text{OD}})$ must have the same size, which means that $\vec{u}(q^{\text{OD}})$ is on the boundary of the circle with radius 0.5 centered at $(0, 0.5)$, as shown in Figure 22. By Corollary 2, type 1's alternative a_1 is $a(q)$ for all $q \leq q^{\text{OD}}$, and $a_1 = a(q^{\text{OD}})$ for all $q > q^{\text{OD}}$ (since $a^{\text{BD1}} = a(q^{\text{OD}})$ is the unique element in $B_1(q^{\text{OD}}) \cap A^=$).

The second case is when $\vec{u}(1)$ lies in region 3.b. In this case, as shown in Figure 23, the line that connects $\vec{u}(2)$ to $\vec{u}(1)$ touches the circle at some point. This point

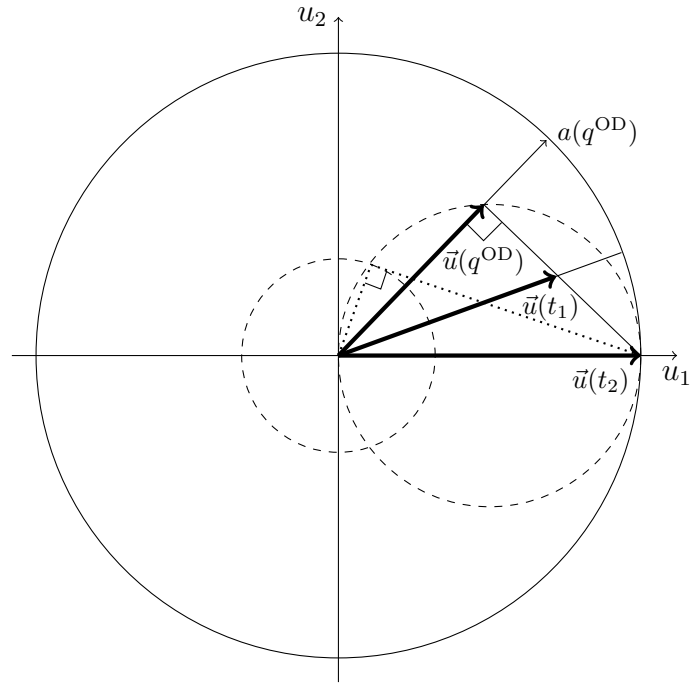


Figure 22: Vector $\vec{u}(t_1)$ is above the tangent from $\vec{u}(t_1)$ to the circle with radius c centered at the origin. Threshold q^{OD} is such that $\vec{u}(q^{\text{OD}})$ is on the boundary of the circle with radius 0.5 centered at $(0, 0.5)$.

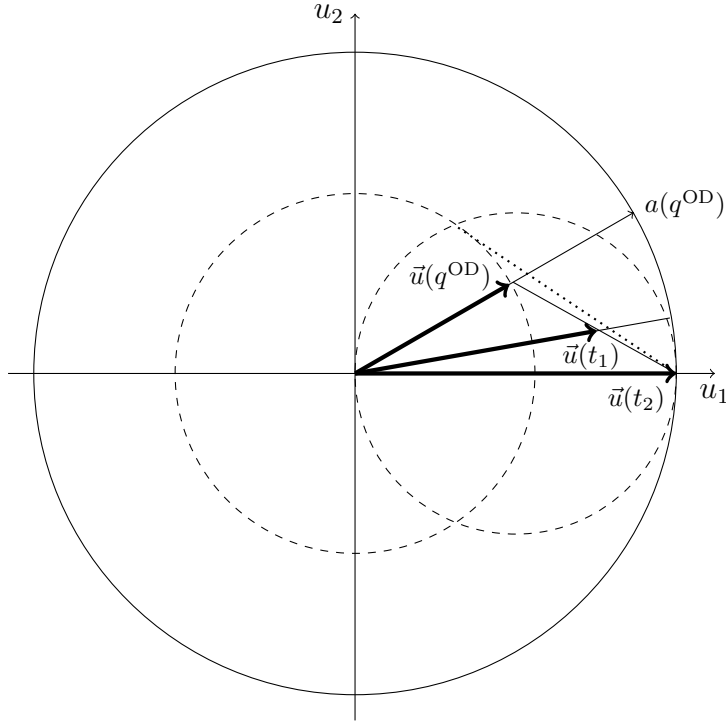


Figure 23: Vector $\vec{u}(t_1)$ is below the tangent from $\vec{u}(t_1)$ to the circle with radius c . The threshold q^{OD} is such $\vec{u}(q^{\text{OD}})$ has size c .

corresponds to q^{OD} . Indeed, for $q < q^{\text{OD}}$, we have that $s(\vec{u}(q)) > c$, so $B_1(q^{\text{OD}})$ is the single alternative $a(q)$ corresponding to the unit vector in the direction of $\vec{u}(q)$. Since the projection of $\vec{u}(2)$ onto the unit vector in the direction of this vector $\vec{u}(q)$ is larger than the projection of $\vec{u}(1)$ onto this vector, we have that $v_2(a(q)) > v_1(a(q))$. For $q > q^{\text{OD}}$, we have that $s(\vec{u}(q)) < c$, so the single alternative in $B_1(q)$ is the outside option, for which both types have valuation 0. Thus, q^{OD} given by (OD) is such that $\vec{u}(q)$ touches the circle with radius c , so $s(\vec{u}(q^{\text{OD}})) = c$ and $B_1(q^{\text{OD}})$ consists of the outside option and all the vectors in the direction of $\vec{u}(q^{\text{OD}})$, for which type 2's valuation is weakly higher than type 1's. By Corollary 2, type 1's alternative a_1 is $a(q)$ (the unit vector in the direction of $\vec{u}(q^{\text{OD}})$) for all $q < q^{\text{OD}}$, a_1 is the outside option for all $q > q^{\text{OD}}$, and can be the outside option or any vector in the direction of $\vec{u}(q^{\text{OD}})$ for $q = q^{\text{OD}}$.

I Interpretation of Giving a Product with Negative Valuation

Recall that valuations in our setting are normalized and represent willingness-to-pay net of costs. To interpret what “giving a product with negative valuation” means in the pre-normalized setting, consider two cases. First, suppose every type has a positive willingness-to-pay (so products are “goods”) but costs might be higher than willingness-to-pay so normalized valuations are negative. Then the seller might sell a product for which both normalized valuations are negative, even though he is “losing money” on that product because it costs more to produce them than either type is willing to pay. Second, suppose some type might have a negative willingness-to-pay for a product (so some products are “bads”), in which case we have to distinguish the cases where free disposal is possible and when it is not (free disposal might be natural with pre-mixed products such as nutritional supplements or assembled devices, but not natural in some settings such as a cable TV subscription). Without free disposal, the interpretation is the same: products might be assigned even if it costs more to produce them than either type is willing to pay. With free disposal, if a type’s willingness to pay for a product is negative, then we can replace it with zero and assume no-free-disposal without loss. But then, if both types have negative willingness-to-pay for a product, those valuations are replaced with zero. Then with positive costs, our result says the seller would never find it optimal to assign such a product.

J Bundling Characterization with Any Number of Products

If both types are uncontested, then the first-best is implementable. So, for the rest of the discussion, assume that type 2 is uncontested (Assumption 1), as we did in Theorem 1, and that type 1 is contested, so $v_2(a_1^E) > v_1(a_1^E)$. This assumption means that

$$\sum_{i: v_1(i) > 0} (v_2(i) - v_1(i)) > 0.$$

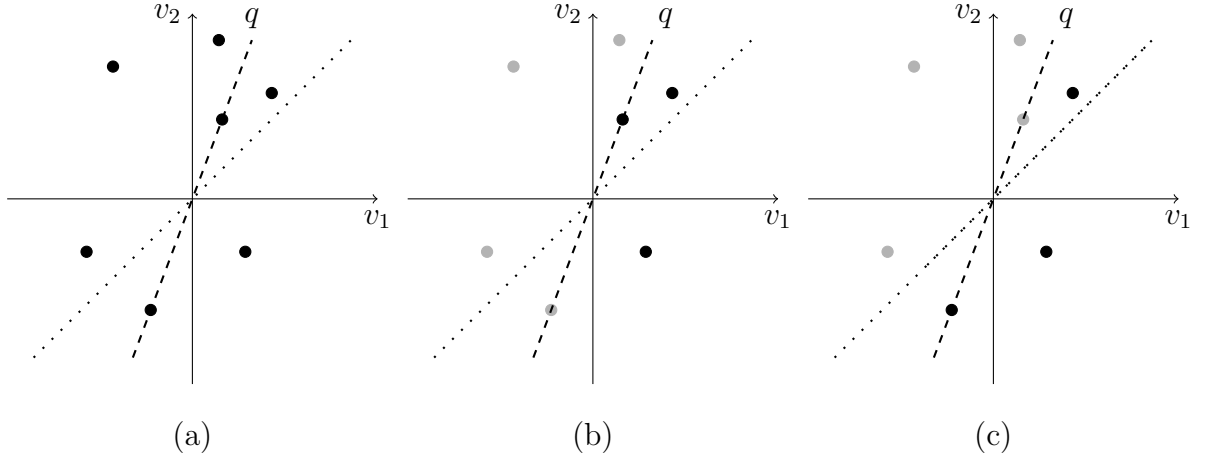


Figure 24: (a) A bundle in $B_1(q)$ contains the products to the right of the hyperplane q and may or may not contains the products on the hyperplane. (b) $a^{1Q}(q)$ contains the products shaded dark. (c) $a^{3Q}(q)$ contains the products shaded dark.

We characterize the optimal random menus by using Corollary 3. An optimal menu is pinned down by type 1's random alternative x_1 . To characterize this random alternative, we identify the set

$$B_1(q) = \operatorname{argmax}_{a \in A} v_1(a) - qv_2(a) = \operatorname{argmax}_{a \in A} \sum_{i \in a} (v_1(i) - qv_2(i)).$$

By definition, a bundle a is in $B_1(q)$ if and only if it contains every product i for which $v_1(i) - qv_2(i) > 0$ and contains no product i for which $v_1(i) - qv_2(i) < 0$, that is,

$$\{i \mid v_1(i) - qv_2(i) > 0\} \subseteq a \subseteq \{i \mid v_1(i) - qv_2(i) \geq 0\}.$$

Each set of products in $B_1(q)$ can be visualized as a bundle that contains every product to the right of the hyperplane that goes through the set of points (v_1, v_2) such that $v_1 - qv_2 = 0$, and may or may not contain the products on the hyperplane, as shown in Figure 24, (a).

Corollary 3 says that there exists a threshold q^{CD} such that $B_1(q^{\text{CD}}) \cap A^+$ and $B_1(q^{\text{CD}}) \cap A^-$ are non-empty. That is, there are two alternatives $a, a' \in B_1(q^{\text{CD}})$ such

that

$$\sum_{i \in a} v_2(i) - v_1(i) \geq 0, \text{ and, } \sum_{i \in a'} v_2(i) - v_1(i) \leq 0.$$

Corollary 3 singles out the alternatives in $B_1(q^{\text{CD}})$ that maximize or minimize $v_2 - v_1$. These alternatives are easy to characterize in our application. The alternative that maximizes $v_2 - v_1$ among those in $B_1(q)$ is the bundle that, among the products i that are on the hyperplane $v_1 - qv_2 = 0$, contains exactly those for which $v_2(i) - v_1(i)$ is positive. Because $0 < q < 1$, those are the products that are in the first quadrant, as shown in Figure 24, (b). Formally, the maximizer is the bundle

$$a^{1Q}(q) = \left\{ i \mid \left(v_1(i) - qv_2(i) > 0 \right) \text{ or } \left(v_1(i) - qv_2(i) \geq 0, v_1(i), v_2(i) \geq 0 \right) \right\}. \quad (6)$$

Similarly, the minimizer of $v_2 - v_1$ among those in $B_1(q)$ is the bundle that contains the products on the hyperplane that are in the third quadrant, as shown in Figure 24, (c), that is,

$$a^{3Q}(q) = \left\{ i \mid \left(v_1(i) - qv_2(i) > 0 \right) \text{ or } \left(v_1(i) - qv_2(i) \geq 0, v_1(i), v_2(i) \leq 0 \right) \right\}. \quad (7)$$

Given the above definitions, we state our characterization result as follows.

Proposition 2 *Suppose that randomization is allowed and $\sum_{i: v_1(i) > 0} (v_2(i) - v_1(i)) > 0$. For any probability $q \in (0, 1)$ of type 2, a menu $\{(x_1, p_1), (x_2, p_2)\}$ is optimal if and only if it satisfies the following properties:*

1. *Type 2's allocation is efficient ($x_2 = a_2^E = \{i \mid v_2(i) > 0\}$).*
2. *Type 1's individual rationality constraint IR1 holds as an equality:*

$$p_1 = v_1(x_1).$$

3. *Type 1's allocation and type 2's binding constraints are specified given q as follows:*

- (a) *If $q < q^{\text{CD}}$, then x_1 is any randomized bundle that contains every product i with $v_1(i) - qv_2(i) > 0$ with probability 1 and every product i with $v_1(i) - qv_2(i) < 0$ with probability 0, and IC2 holds as an equality.*

- (b) If $q = q^{\text{CD}}$, then x_1 is any randomized bundle that contains every product i with $v_1(i) - qv_2(i) > 0$ with probability 1 and every product i with $v_1(i) - qv_2(i) < 0$ with probability 0 and such that

$$E_{a \sim x_1}[v_2(a) - v_1(a)] \geq 0$$

and IC2 holds as an equality. If the above inequality holds as an equality, then additionally IR2 holds as an equality.

One particular x_1 that satisfies the above inequality assigns probability 1 to bundle $a^{1Q}(q^{\text{CD}})$ defined in (6).

- (c) If $q > q^{\text{CD}}$, then x_1 is any randomized bundle that contains every product i with $v_1(i) - qv_2(i) > 0$ with probability 1 and every product i with $v_1(i) - qv_2(i) < 0$ with probability 0 and such that

$$E_{a \sim x_1}[v_2(a) - v_1(a)] = 0$$

and IR2 holds as an equality.

One particular x_1 that satisfies the above equality randomizes only over the two bundles $a^{1Q}(q^{\text{CD}})$ and $a^{3Q}(q^{\text{CD}})$, defined in (6) and (7).

One possibility for generating the alternative x_1 in property 3(c) of Proposition 2 is to randomize only over the two alternatives $a^{3Q}(q^{\text{CD}})$ and $a^{1Q}(q^{\text{CD}})$. In the resulting random alternative, products to the right of the hyperplane $v_1 - qv_2 = 0$ are included with probability 1, each product on the hyperplane in the first quadrant is included with some probability p , and each product on the hyperplane in the third quadrant is included with the complementary probability $1 - p$. Except for knife edge cases, however, there is a single product i on the hyperplane. The resulting random alternative x_1 then consists of all the products to the right of the hyperplane along with product i with some probability.

K A Bundling Example when Randomization is not Allowed

We here return to the example from Section 5.2 and describe the optimal menus when randomization is not allowed. For convenience, Figure 25 recreates the figures in the

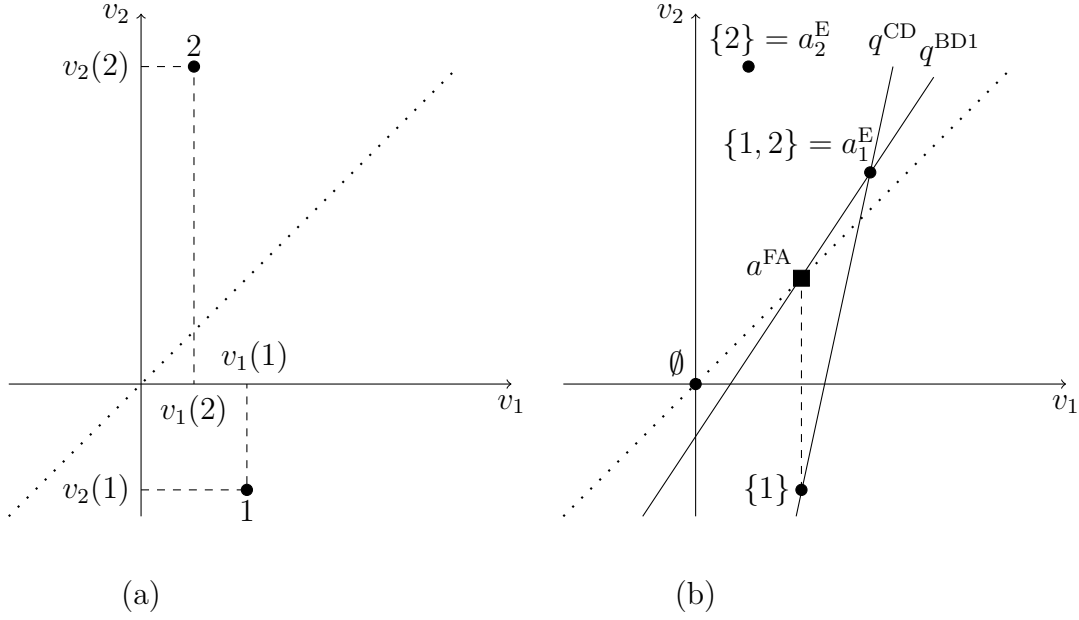


Figure 25: (a) The product space. (b) The alternative space with thresholds q^{BD1} and q^{CD} .

product space and the alternative space from Section 5.2.

We identify the optimal menus using Theorem 1. For this, we need to identify the threshold q^{BD1} , which is the lowest q at which the fictitious alternative a^{FA} is maximal in the direction of the vector $(1, -q)$ among the alternatives $A^+ \cup \{a^{FA}\}$. In our example, A^+ consists of alternatives $\{2\}$ and $\{1, 2\}$, and the threshold q^{BD1} is such that alternative $\{1, 2\}$ and the fictitious alternative a^{FA} lie on a hyperplane orthogonal to the vector $(1, -q^{BD1})$, so

$$v_1(\{1, 2\}) - q^{BD1}v_2(\{1, 2\}) = v_1(a^{FA}) - q^{BD1}v_2(a^{FA}).$$

This is shown in Figure 25, (b). For any $q < q^{BD1}$, alternative $\{1, 2\}$ is maximal in the direction of the vector $(1, -q)$ among the alternatives $A^+ \cup \{a^{FA}\}$, and for any $q > q^{BD1}$, a^{FA} is maximal in the direction of that vector. Recall that q^{CD} is the threshold at which $\{1, 2\}$ and a^{BD1} are maximal (among all alternatives) in the direction of the vector $(1, -q^{CD})$, and notice that $q^{BD1} > q^{CD}$, as shown in Figure 25, (b).

We now describe type 1's alternative in any optimal (non-random) menu. If $q < q^{BD1}$, then type 1's alternative is the grand bundle $\{1, 2\}$. If $q = q^{BD1}$, then

type 1's alternative can be either $\{1, 2\}$ or $a^{\text{BD1}} = \{1\}$. If $q > q^{\text{BD1}}$, then type 1's alternative is $a^{\text{BD1}} = \{1\}$.

There is a connection between the optimal menus with and without randomization. For $q > q^{\text{BD1}} > q^{\text{CD}}$, with randomization type 1's alternative is a distribution over $\{1, 2\}$ and $\{1\}$; without randomization type 1's alternative is $a^{\text{BD1}} = \{1\}$. One might wonder if this is a general property: is it always the case that a^{BD1} , which is type 1's alternative in the optimal deterministic menu for any $q > q^{\text{BD1}}$, is also in the support of type 1's random alternative in an optimal random menu for $q > q^{\text{CD}}$? This property is not in fact general. In the example, $a^{\text{BD1}} \in B_1(q^{\text{CD}})$ and, moreover, it is the only alternative in $B_1(q^{\text{CD}})$ that lies below the diagonal in the alternative space, so it has to be used in type 1's random alternative for any $q > q^{\text{CD}}$. We now give an example in which $a^{\text{BD1}} \notin B_1(q^{\text{CD}})$, so a^{BD1} is not in the support of type 1's random alternative in the optimal random menu for any $q > q^{\text{CD}}$, but a^{BD1} is type 1's alternative in the optimal deterministic menu for any $q > q^{\text{BD1}}$.

For the example, we add a third product, product 3, without changing the other two products. The result in the product space is shown in Figure 26, (a). To depict the result in the alternative space, we take each of the four existing alternatives and move it in the direction of the vector $(v_1(3), v_2(3))$ to obtain four new alternatives, each containing product 3, for a total of eight alternatives. Both types have a negative valuation for product 3, so the efficient alternatives remain unchanged. For certain parameters, the threshold q^{CD} and the set $B_1(q^{\text{CD}}) = \{\{1\}, \{1, 2\}\}$ remain unchanged.⁴² Importantly, $B_1(q^{\text{CD}})$ does not contain the bundle $a^{\text{BD1}} = \{1, 2, 3\}$. Thus, for $q > q^{\text{BD1}}$ type 1's alternative in the optimal deterministic menu is the bundle $\{1, 2, 3\}$, but in the optimal random menu $\{1, 2, 3\}$ is not in the support of type 1's random alternative, which randomizes over bundles $\{1\}$ and $\{1, 2\}$.

⁴²Specific valuations that work are $v_1(1) = 2$, $v_1(2) = 1$, $v_1(3) = -0.5$, $v_2(1) = -2$, $v_2(2) = 6$, and $v_2(3) = -2$.

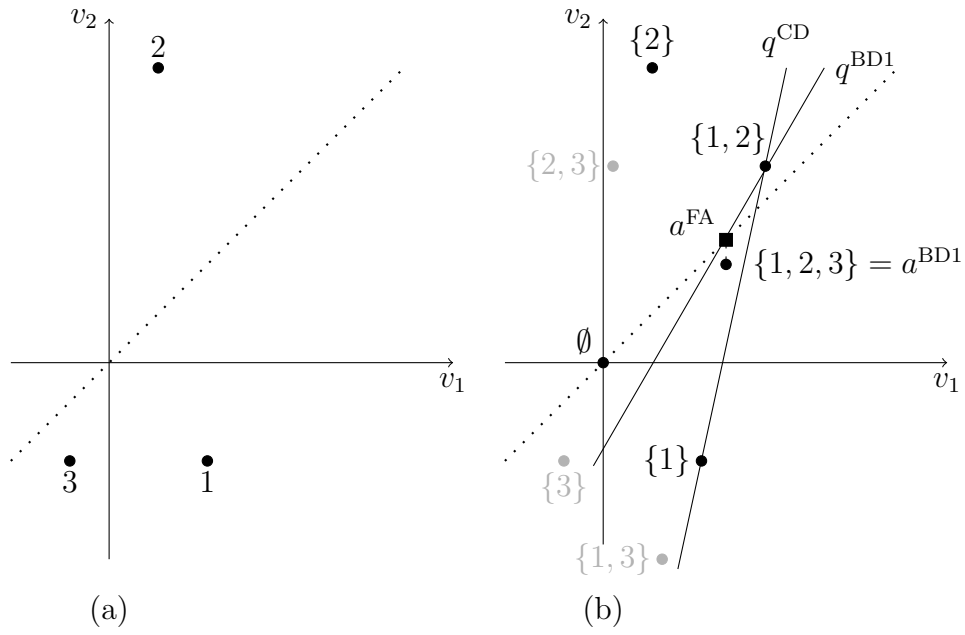


Figure 26: (a) The product space with three products. (b) The corresponding alternative space with eight alternatives. Alternative that are not important in the example are lightly shaded to ease visualization.